

NEXTGENERATION



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Note from the Editor

Consider that a mission to the Moon, nuclear power, and the Internet were each once thought of as heralding a new generation. Today, we live in that generation. We think instead about the impact of bigger and better ideas, like plans for a human colony on Mars, or the capacity to genotype thousands of consumers curious about their DNA. We are witnesses also to smaller and savvier advances: proteins and viruses that can reach intracellular targets with astonishing precision; smartphones, smart watches, and even smart fridges.

All of these topics and more are explored in the following pages, in which our writers detail not only the marvels of accelerated scientific progress, but also the unavoidable ethical and logistic dilemmas brought by such progress. In our interviews with three Berkeley professors, we ask about the implications of their pioneering work in the distinct fields of psychology, ecology, and neurobiology. Notably, as undergraduate students at UC Berkeley, we are lucky to attend a school that houses so many labs powering next-generation research.

What is considered 'next-generation' has always been a relative concept—an event or advancement that feels as if it could belong in another lifetime. By exploring today's 'futuristic' areas of science, this issue, I hope, will make them seem more familiar.

—Rachel Lew, Editor-in-Chief



Table of Contents

Features

Cancer Drugs: Targeting Undruggable Proteins (Yana Petri).....	pg.1
Drug Delivery: Steps Toward Replacing the Pill (Shivali Baveja).....	pg.4
The Road to the Red Planet (Matt Lundy).....	pg.7
The Future of Privacy and Cybersecurity (Jennifer Zeng).....	pg.10
Tackling Complex Diseases with Epistasis (Kara Jia).....	pg.13
Drawing the Line between Predators and Livestock (Lara Volski).....	pg.16
Going Public with Personal Genetics (Marie Balfour).....	pg.19
Crossing the Synaptic Cleft: Treating Autism Spectrum Disorder (Sanika Ganesh).....	pg.23
The Future of Work (Macy Chang).....	pg.27
Powering the Future with Hydrogen Fuel Cells (Mina Nakatani).....	pg.30
Nanomedicine and Its Various Applications (Nicole Xu).....	pg.33

Interviews

Shifting Power Dynamics: The #MeToo Movement (Arjun Chandran, Cassidy Hardin, Michelle Lee, Melanie Russo, & Yana Petri).....	pg.35
Explaining Patterns in Ecology: Climate Manipulation and Mathematical Modeling (Nikhil Chari, Rosa Lee, Phillip de Lorimier, Moe Mijjum, Sona Trika, & Elena Slobodyanyuk).....	pg.39
Adeno-Associated Viruses: Vehicles for Retinal Gene Therapy (Arjun Chandran, Cassidy Hardin, Michelle Lee, Melanie Russo, & Elena Slobodyanyuk).....	pg.44
Tips on Science Careers: Your Graduate Student Instructors Share Wisdom (Arjun Chandran, Nihikil Chari, Cassidy Hardin, Michelle Lee, Rosa Lee, Phillip de Lorimier, Moe Mijjum, Melanie Russo, & Sona Trika).....	pg.49

Research

The Mysterious Loss of the Third Molar in the New World Monkey Family <i>Callitrichidae</i> and its Relationship to Phyletic Dwarfism (Jeffrey L. Coleman).....	pg.53
Assessment on Interoperability of Health Information Exchanges (Varun Neil Wadhwa).....	pg.60



CANCER DRUGS: TARGETING UNDRUGGABLE PROTEINS

BY YANA PETRI

Many of us know someone who has been affected by cancer. At first glance, current cancer statistics look grim: the average American has about a 40% lifetime risk of developing cancer and a 20% chance of dying from it.¹ Yet there is reason to be optimistic. In the past decade, scientists have tremendously advanced our understanding of underlying causes of cancer, while also developing novel drugs that may potentially deliver safer and more robust cancer treatments than conventional FDA-approved small-molecule inhibitors.

Many prominent scientists in the field of drug discovery, including Brent Stockwell, a chemical biologist at Columbia University in New York, believe that the challenge of coming up with new cancer treatments lies in undruggable disease-causing proteins.^{2,3} These mutated or aberrantly regulated proteins are considered extremely difficult to target with available drug discovery technologies.³ A poster child of undruggable

proteins is mutated K-Ras, which is responsible for about one-third of all cancers.³

Ras protein family members (K-Ras, N-Ras, and H-Ras) have been studied for more than thirty years and are thought to play a crucial role in the regulation of cell proliferation, differentiation, and survival, by signaling through a number of downstream pathways.⁴ The signaling function of Ras proteins is tightly controlled by the cell through a mechanism that resembles a light switch. In healthy cells, Ras proteins cycle between a GDP-bound state—when the growth signal is off—and a GTP-bound state—when the growth signal is on.³ The transition between these two states is in part regulated by binding of GTPase-activating proteins (GAPs). These vigilant regulators convert GTP to GDP and make sure that the cell growth signaling function of Ras proteins gets turned off in a timely manner. Mutations in genes that code for the structure of Ras proteins can interfere with GAP binding. This causes Ras pro-

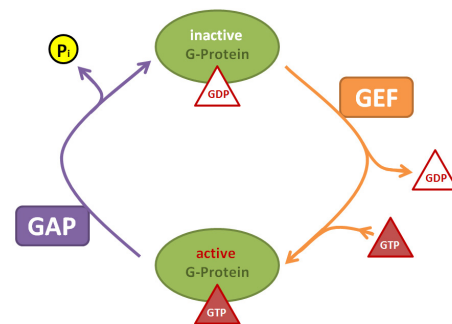


Figure 1. Ras (a type of G-protein) cycle that shows the GDP-bound “off” state and GTP-bound “on” state.

teins to get stuck in a permanently activated “on” state, which ultimately leads to uncontrolled proliferation of Ras-mutant cells.⁴ Scientists are not yet sure what causes mutations in Ras genes, but the evidence that these mutations contribute to cancer is indisputable.³

But what makes K-Ras and other undruggable proteins so elusive? Most small-molecule drugs are inhibitors: they work by binding to a protein and physically blocking its function.³ It is helpful to think of a drug as a key and of the protein as a lock. In order to bind, the drug must fit snugly into a pocket on the surface of a protein. In fact, the presence of well-defined pockets is the hallmark of druggable proteins.² Undruggable proteins like K-Ras are quite different—they do not have obvious binding pockets. Some are annoyingly smooth; some are floppy and disordered. Others like to form strong interactions with nearby proteins and evade small molecules that attempt to separate them. Astonishingly, about 90% of all proteins are considered undruggable.⁵ Because of this challenge, drugs that were approved by the FDA before 2009 collectively interacted with just 2% of all proteins in human cells.⁵

With these considerations in mind, what is the future of cancer drug discovery? Kevan Shokat and Craig Crews—two leading researchers in chemical biology—have emphasized the need for creative chemical approaches to developing novel cancer drugs. And while it is unclear which approach will ultimately cure cancer, one thing is certain: novel inhibitors and proteolysis targeting

chimeras (PROTAC) are beginning to generate a lot of excitement in the field.

In 2013, Shokat and his team discovered that a specific type of mutant K-Ras, K-Ras(G12C), has a druggable surface pocket that was not apparent in previous crystallographic studies.⁷ To come up with a starting structure of a novel K-Ras inhibitor, the scientists screened a library of small molecules against K-Ras and used mass spectroscopy to identify compounds that were able to bind to the newly discovered pocket. Next, they obtained a 3D crystal structure of K-Ras bound to the most promising compound. Based on this structural data, Shokat and his team optimized the lead compound and came up with a first novel inhibitor of K-Ras.

Many follow-up studies came out after Shokat's discovery, but some researchers doubted that the new pocket would be druggable in living organisms. However, in 2018, Yi Liu and others reported that tumors indeed decreased in mice that were treated with the new K-Ras inhibitors.⁸ This study served as a key step towards starting clinical trials with patients who have a specific K-Ras mutation.⁸ In parallel to Shokat, Crews developed another strategy to target undruggable proteins. His lab came up with “smart” small molecules—proteolysis-targeting chimeras, or PROTACs for short—that can entirely de-

“...about 90% of all proteins are considered undruggable.”

stroy proteins instead of just blocking their function.⁹

The mechanism through which PROTACs induce protein degradation is beautiful and simple. PROTACs consist of three components: the “head,” which binds the cancer-causing protein, the linker, and the “tail,” which binds another protein called E3 ligase.⁹ The E3 ligase attaches ubiquitin to the cancer-causing protein, which acts as a signal for the activation of the natural cellular quality-control machinery. In this mechanism, cancer-causing proteins tagged with a ubiquitin chain are recognized by the proteasome, which literally rips them to pieces.

PROTACs have many advantages over traditional inhibitors. They have long-lasting effects, are likely safer, and do not induce drug resistance.⁹ However, their biggest benefit lies in the ability to truly “drug the undruggable.” Because PROTACs can in theory bind anywhere on the cancer-causing protein—for instance, to a surface pocket that is not involved in protein function—all cancer-causing proteins, including those that don't have drug-binding pockets, have the potential to be destroyed.

PROTACs are already poised to enter the mainstream. A couple of years ago, Crews launched a startup called Arvinas,

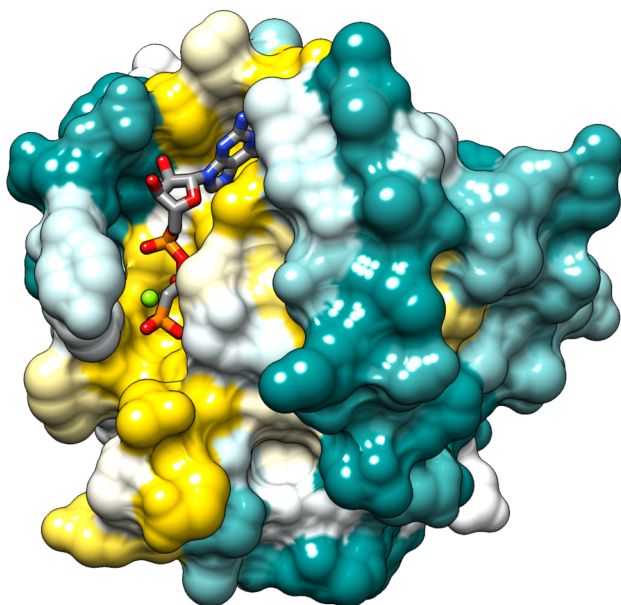


Figure 2. Surface structure of K-Ras analogue H-Ras bound to GDP. Shokat's team discovered a new pocket on its surface that was not apparent in previous crystallographic studies.⁷

“...not a single drug has been FDA-approved against mutant protein K-Ras, which is responsible for about one third of all cancers.”

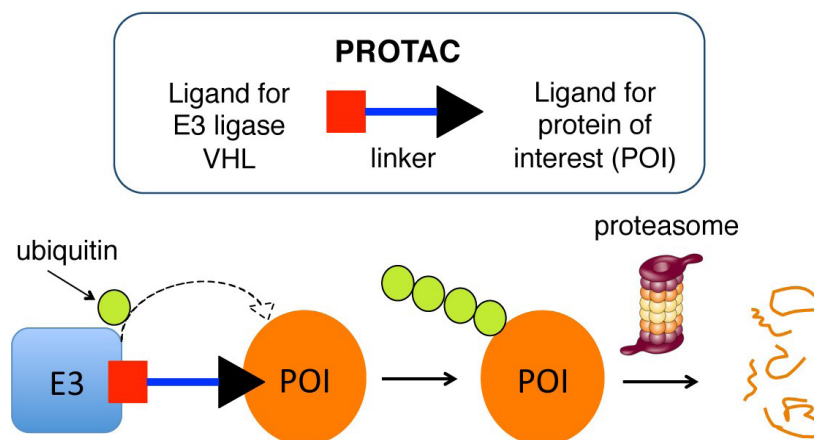


Figure 3. Schematic of how PROTACs degrade cancer-causing proteins.

which openly announced its plans to take PROTACs into clinical trials in the near future. Several labs have also recently used PROTACs to degrade a challenging protein target named BRD4, which is important in the onset of leukemia.¹⁰

Stephen Hawking once said: “I believe things cannot make themselves impossible.” Indeed, it is up to us to create new technologies that render the term “undruggable” obsolete. While this vision currently lies just out of reach, such advances in drug development allow us to imagine a not-too-distant future of personalized medicine in which patients can be cured by taking drugs that target the unique mutations that underlie their cancer.

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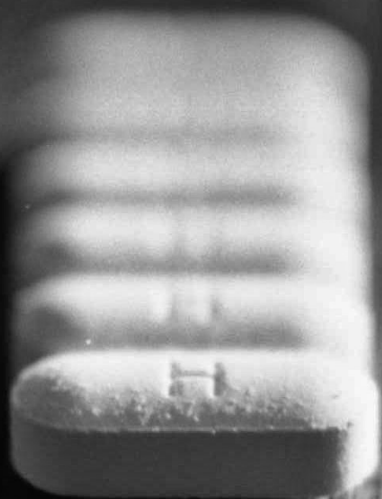
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DRUG DELIVERY: STEPS TOWARD REPLACING THE PILL

BY SHIVALI BAVEJA



Be it through your bloodstream or the lining of your intestinal tract, the human body uses a multitude of pathways to deliver molecules to their appropriate destinations. For example, the aspirin you took to dull your splitting headache travels through multiple networks in order to inhibit the enzymes that propagate your pain. Drug delivery, the process of getting medications to their intended targets, is a complex and constantly evolving field with a goal of making treatments faster, safer, and more effective. With the advent of new tools and techniques such as biodegradable microspheres and ultrasound-mediated delivery, current developments in biotechnology offer a promising outlook on simplifying and improving patients' relationships with their medication.

The majority of research in the field today centers on finding new ways to localize drug delivery. By preventing medication from deviating from its intended target,

some of the negative consequences of drug interactions can be minimized. One of the most rapidly growing areas of development is that of transdermal drug delivery, or drug delivery through the skin. By passing molecules painlessly through the skin, this form of delivery tends to be more efficient as it requires lower doses of medication. The development of biosensors that measure the release of drugs has further allowed transdermal delivery systems to closely monitor the intake of molecules.² This brings us to one of the greatest advantages of drug delivery: patient-specific treatment plans. By maintaining closed-loop systems which have the ability to control the rate of medications delivered, treatments can be tailored not only to the patient as a whole, but also to variations in their daily life.⁴

Transdermal patches offer further advantages in their ease of delivery as they do not have to be ingested or injected. Since the patches require little effort to use, they

are expected to significantly increase levels of patient compliance.⁸ Behind the scenes of transdermal delivery, techniques for electroporation are at work. Electroporation, a method of administering medications using extremely small electrical impulses, makes it easier for water-insoluble medications to make it past the protective barrier of the skin.⁹ This technique reduces the resistance molecules face and is necessary for the permeation of medications inside the bloodstream and body. In instances where transdermal delivery is not feasible, other alternative techniques are in development.

Conditions such as macular degeneration, which affect the eye, are instances where patches no longer are a viable treatment option. Instead, researchers are experimenting with biodegradable microspheres. These microspheres are essentially little bubbles carrying drugs which "pop," or break down over time, and release their enclosed medications at a steady rate. As a

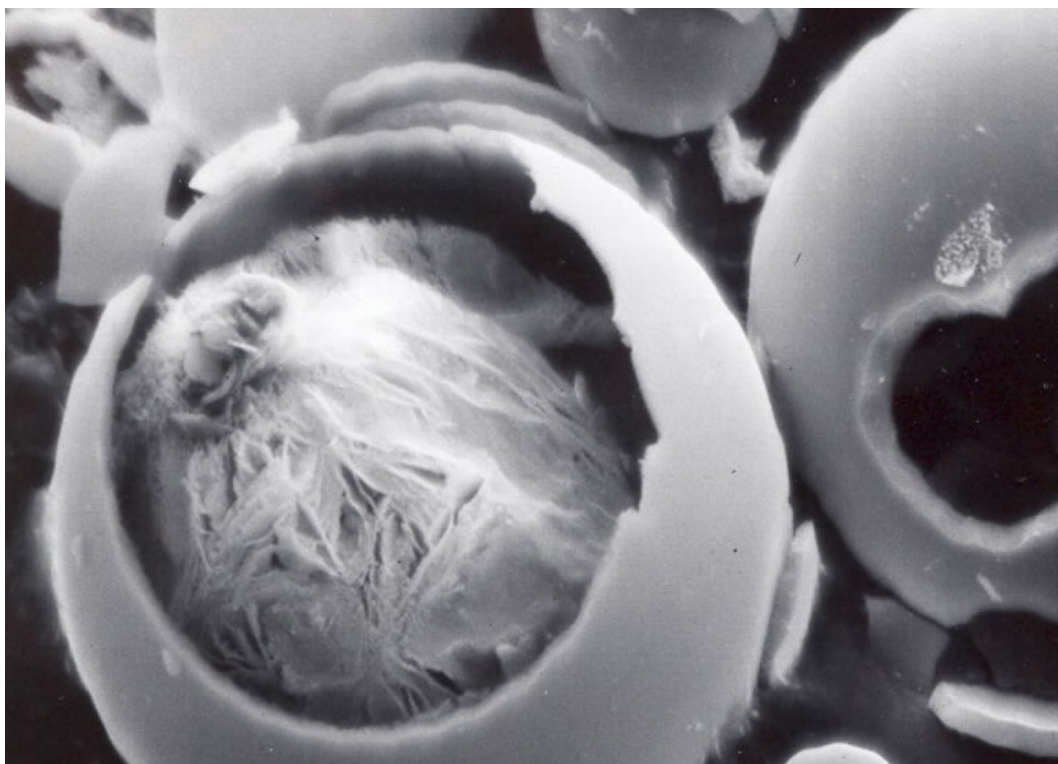


Figure 1. Recent developments with glass biodegradable microspheres have allowed for the delivery of drugs in small controlled quantities that can be released over regular periods of time.¹¹

result, this technique offers a means for sustained delivery with less frequent intervention.¹ Ultrasound-mediated drug delivery, another technique utilizing bubbles of medication, uses high frequency waves to release the medications as opposed to having them break down on their own. The use of a closed-loop feedback system also offers the possibility of bursting bubbles when a relevant

stimulus in the body leads to “popping” the bubbles.⁵ *In vivo* results of the use of ultrasound during application with swines’ gastrointestinal tracts has resulted in successful dispersal of medication and is a promising indication of future application in humans.¹⁰

Just as innovation erupts in the field of delivery, tools to refine these mechanisms develop alongside it. The process of delivering medications relies on two distinct components: the molecules themselves and the mechanisms of their entry and regulation. Developments in the field of microfluidics have led to the design of micro- and nano-particles that have been engineered to affect only localized areas in the body. One of the greatest advantages of microfluidics is its ability to model biological systems, a tool which is helpful in *in vitro* models that simulate an organism’s response to and interaction with a substance. Thus, this tool is likely to be integrated into *in vitro* drug delivery models in the development of medications and transport systems to effectively speed up the delivery of micro- and nano-particles in the body.⁶

Another tool being utilized in the specialization of the drug delivery process is mi-

crochips. The precision of these small computers allows doctors to walk the fine line between doses that are too low to be effective and those which are high enough to be toxic to patients. Microchip technology can be applied to improve treatments for cancer and other conditions such as osteoporosis, which require regular feedback-based delivery of medications.³ Clinical studies on the healing process with microchips have shown that minimal disturbances occurred after implantation, and this is promising regarding their use in future drug delivery mechanisms. To this extent, prior *in vitro* studies have corroborated these findings.³

Despite knowing that many tools and techniques are being developed, it is still important to recognize the obstacles which some of these strategies will likely face in the future. First, the patient-specific nature of many of these models makes them very time- and labor-intensive to develop and implement. As a result, the feasibility of such techniques on the larger scale may be diminished, and use in the mass market may take much more time and optimization.² Additionally, continued research is necessary to investigate the long-term ef-

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"In vivo results of the use of ultrasound during application with swines' gastrointestinal tracts has resulted in successful dispersal of medication and is a promising indication of future application in humans."

fects of some carrier techniques such as those in biodegradable microspheres and ultrasound-mediated delivery. Further clinical studies on such mechanisms must occur before conclusions can be drawn about their impacts on human health.^{7,9} It is only after clinical studies are underway that we can hope to see such revolutionary technology make it into the hands of patients. Finally, even with the technology we have already developed, these methods must be perfected *in vivo* before they can be effectively translated to the healthcare setting.¹⁰

Based on today's potential avenues for drug delivery, such techniques offer enormous scope for revolutionizing the way medications are administered. If the current research is soon translated to further clinical trials, such tools will likely reach everyday consumers within the next decade and forever alter the relationship between individuals and their drugs. Perhaps in a few years from now, pill bottles will have permanently become a thing of the past

and individuals will be taking lower doses of drugs to treat their conditions. Until then, the field of drug delivery will continue progressing to make this future a reality.

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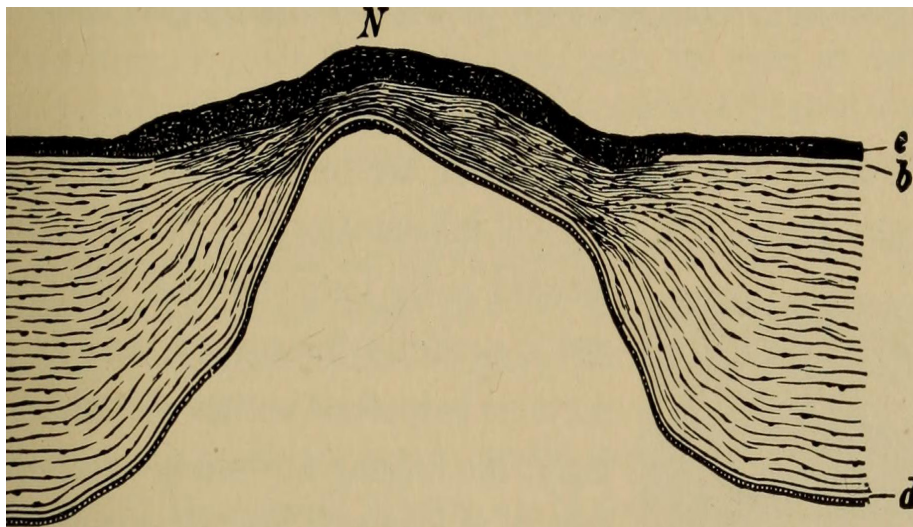


Figure 2. The cornea as visualized here reveals the membrane through which biodegradable microspheres must be delivered.¹



BY MATT LUNDY

The Road to the Red Planet



For humanity to secure its survival in the unforeseen future, we must go to Mars. At least, that's what Elon Musk, leader of space exploration company SpaceX, believes. Indeed, if humanity were to become a multi-planet species, it would increase its chances of long-term survival. However, establishing a self-sustaining colony on Mars would be one of the greatest challenges humanity has ever faced. There are a number of potential issues that must be addressed, from technological and economical concerns to psychological and political ones. Nevertheless, the advantages behind such an undertaking may be too great to forfeit.

There are at least two incentives for a long-term human settlement on Mars. First, an interplanetary species has immensely higher odds of long-term survival. If we were to live on two (or more) planets, we would have a backup in case of doomsday events such as super volcano eruption, giant asteroid impact, nuclear holocaust, or resource depletion. As Musk argues,

"One path is we stay on Earth forever, and then there will be some eventual extinction event... The alternative is to become a space-bearing civilization and a multi-planetary species, which I hope you would agree is the right way to go."⁶ In choosing to become interplanetary, humanity would be declaring its will to survive, even in the face of possibly apocalyptic circumstance.

From a more grounded perspective, the technological advancements that would result from pushing humans deeper into space are also immense. Past space ventures have generated vital technologies ranging from LEDs to solar energy acquisition and water purification advancements.¹¹ The strict and severe parameters that accompany space travel push engineers working on such projects to innovate in ways unseen in earthly projects. A Martian colony would be the greatest space undertaking yet, so the advancements that accompany it could be expected to be of similar magnitude.

Even if the motivations of going to Mars

were unquestioned, reaching the Red Planet is still quite a challenge. Most proposed plans at this point share a similar overarching structure. Multiple missions of people and rockets would be sent to Mars in relatively quick succession. The ideal colony would be self-sufficient as soon as possible, as that would remove the risk of the

"Even if the motivations of going to Mars were unquestioned, reaching the Red Planet is still quite a challenge."

colony failing due to problems on Earth.⁴ This means that the colony would need enough people and materials to operate independently, from building replacement parts and gathering its own resources, to even producing its own rocket propellant.⁶ A sustainable colony would then be able to grow into a fully developed civilization and the mission would be a categorical success.

Yet, such a complex and demanding undertaking is sure to be rife with problems. As it stands, a Martian colony would be outrageously expensive. For comparison, the Apollo program cost a staggering \$100-200 billion to send just twelve people to the Moon, despite including no plans of long-term or permanent residence for its astronauts.⁶ Current rocket technology makes a Mars colony unviable for any single party, and unlikely even for a group. Another issue would be the well-being of the colonists. A long spaceflight followed by an entire life spent within the confines of a spacesuit or a close-quarters Martian residence, with severely limited human contact, would be a gargantuan psychological undertaking.² This could then breed larger social problems, such as an in-group/out-group dichotomy, where new colonists would be met with hostility from their seniors who might perceive them as lesser due to their lack of experience.⁴ The legal status of such a colony would also have to be addressed, preferably before the colony's creation. There is very little legislation that concerns property in space; a budding civilization would no doubt have great need for such laws. A final problem to consider would be how to deal with the fatality of prolonged

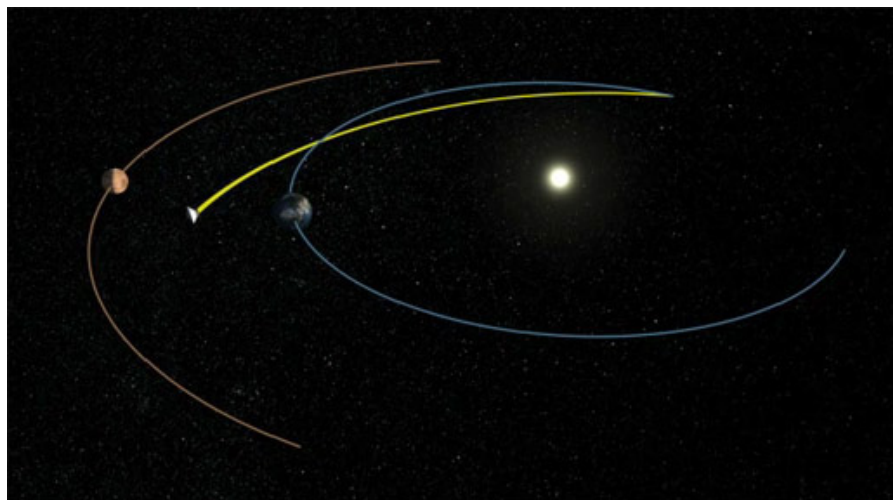


Figure 1. The best time for a mission to Mars would be when the orbits of Earth and Mars are closest, minimizing the distance the rockets have to travel.^{12,13}

dependency on Earth, which would render the benefit of Mars as a backup useless.

Although these are several significant issues, solutions are being theorized, and some are already in production.

In response to the exorbitant cost of a mission to Mars, SpaceX is investing heavily in creating sustainable and reusable rockets. Through technological progressions like this and in-orbit flight refueling, SpaceX estimates (in a best-case scenario) that the price could be reduced to \$100,000-200,000 per trip. While this is expensive, it is quite cost-effective for such a huge mission, especially when compared to the Apollo missions.⁶

To facilitate a self-sustaining colony, initial missions could be outfitted with the extra material needed to construct infrastructure and repair machinery, something that would be made possible by lower costs.

Regarding further enabling self-sufficiency, the SAM (sample analysis) instrument on NASA's *Curiosity* rover found sufficient amounts of reduced sulfur to envisage sulfur redox chemistry as an energy source for supporting life on Mars. In other words, sulfur on Mars can undergo a chemical reaction that would release energy which could then be captured and used to support the energy needs of our potential colony.⁸

Unfortunately, there are not yet many unique solutions to indirect problems such as the psychological well-being of colonists or the potential political issues of a colony. There is some precedent in extensive training for astronauts and the current setup of the diplomatic International Space Station that we can look to for guidance, but as of now, these are underdeveloped solutions that must be improved upon significantly for an actual colony mission.

“If we were to live on two (or more) planets, we would have a backup in case of doomsday events such as super volcano eruption, giant asteroid impact, nuclear holocaust, or resource depletion.”

However, there is one final motivation to make the voyage to the Red Planet, which once inspired millions of everyday people to earnestly yearn for humans to step foot on the moon. In the words of the man who spoke it into existence 56 years ago, former President John F. Kennedy:

*We choose to go to the Moon!
We choose to go to the Moon in this decade... not because [it is] easy, but because [it is] hard; because that goal will serve to organize and measure the best of our energies and skills, because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win...*¹⁵

We can now find this impulse in our pursuit of traveling to Mars. The mission will be long and difficult. At the current time, we assuredly cannot achieve it, but we are getting closer.

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Figure 2. SpaceX is on the road to reusable rockets with their Falcon Heavy boosters, shown displaying the ability to land safely after a launch.¹⁴

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THE FUTURE OF PRIVACY AND CYBERSECURITY

BY JENNIFER ZENG



“Big Brother is watching you.” But rather than being watched through forced surveillance as George Orwell’s 1984 suggests, today, consumers are slowly conceding their own privacy. Already, smart technologies such as refrigerators and thermostats are used without a second thought. With our current rate of technological progress, it is not unreasonable to imagine a future where nearly every tool we use everyday will be connected to the internet. Although the network that connects all these devices—the Internet of Things (IoT)—is only in its early stages, its increasing prevalence will be one of the main issues of next-generation cybersecurity.

Apart from the direct collection and distribution of information, future privacy concerns will revolve around the ability of companies to obtain new information about consumers that is not given voluntarily.⁷ Information will be generated through predictive analytics, which is the use of statistics and data modeling to

make predictions based on existing data. Data produced through predictive analytics is considered new information; under current informational privacy networks, it is unclear whether the consumer or the analytics company has the right to this data and its dissemination.² This issue gives rise to a new model of privacy that consumers must consider—datafication privacy—beyond the current models of surveillance and data collection privacy.

The negative effects of predictive analytics will only be exacerbated by the information consumers unwittingly provide through the IoT. The IoT consists of all smart devices connected to the internet in a residence, such as refrigerators and speakers. Devices connected through the IoT share consumer behavior patterns and other personal information with each other.⁴ This is of particular concern because of the increasing popularity of these devices, which are touted for their efficiency and personalizability. Yet consumers are

largely unaware of how easily hackable devices on the IoT are, and often consider the information they carry to be benign.

The IoT lacks security for several reasons. Devices have varying authentication methods due to differing environments; because there is no standard protocol for authentication, each device represents a point of vulnerability that could compromise the entire system. The ability of hacked devices in an IoT network to compromise the rest, regardless of any dissimilarity between devices, generates a unique form of vulnerability for such systems. A hacked refrigerator is a threat to a thermostat on the same network because of the general features IoT devices must share in order to communicate. The weak security systems of these devices, and their relatively small size and low-power needs, make IoT devices particularly susceptible to distributed denial of service (DDoS) attacks. A denial of service attack occurs when a computer is overloaded with useless incoming data, so it cannot

“If IoT devices are connected across a blockchain network, the information they share through the network will be cryptographically proofed and secured.”

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receive any other information; a distributed DoS occurs when the packets are being sent to a single computer from a large number of sources. The largest attack on the IoT thus far was by malware that utilized unchanged default passwords for routers and similar devices to instigate DDoS.⁸

One possible security solution utilizes blockchain technology. Blockchain is a secure-by-design decentralized model of information storage based on cryptography, which means that it is not controlled by a single computer or operator. This minimizes its chances of being compromised. All the information held by the model is distributed to every computer connected across a shared network, rather than being stored with a third party. Information is stored in “blocks” connected by secure links, and changing even one piece of information at

any point in the “chain” involves gaining consent from over 50% of the remaining network.¹ If IoT devices are connected across a blockchain network, the information they share through the network will be cryptographically proofed and secured.

Another aspect of blockchain that can be utilized to enhance security is coded contracts. Coded contracts can be used to determine who has access to device software, including patches and updates, as well as who can request service on the device.⁶ These contracts are executed on the blockchain, not by a third party, so they cannot be altered. Unlike a regular contract, in which any of the named parties can break the contract while the rest of the parties follow through, all parties mentioned in a coded contract must fulfill their sides of the agreement or none of the terms of the

contract will be executed on the network.

Artificial intelligence is a second possible solution for security. Deep learning is a subset of machine learning, which in turn is one method of producing AI. Deep learning utilizes information sharing and a computing system that resembles the neural networks in the brain in order to form new connections among information. By analyzing the details of past attacks on a network, deep learning can uncover attack patterns to fix security flaws. Because it is self-learning and requires little supervision, it is more effective and efficient than assigning a person to manually update security protocol every time an attack occurs, especially when the underlying mechanism is too complex for a person to debug. It has been shown that deep learning can be used to detect attacks on IoT networks by utilizing a fog-computing method rather than a cloud-computing method.⁵ While cloud computing relies on a central server far away to complete computational processes, fog computing works on the edge of the “cloud” of devices connected to a server, storing data closer to the device in the network that’s using it. This increases the efficiency of IoT and allows faster deep learning calculations.

While cutting-edge technology can be used to provide secure-by-design systems, it is still humans who generate these technologies and use them. Thus, the ongoing development of IoT has led to a new facet of security as well: non-technical cyber hygiene on the part of the consumer. Theresa Miedema of the University of Toronto defines cyber hygiene as security measures that consumers should use to protect their privacy and devices through the internet.⁴ Because of the increased interaction between consumers and smart devices, bad cyber hygiene can affect not only personal de-

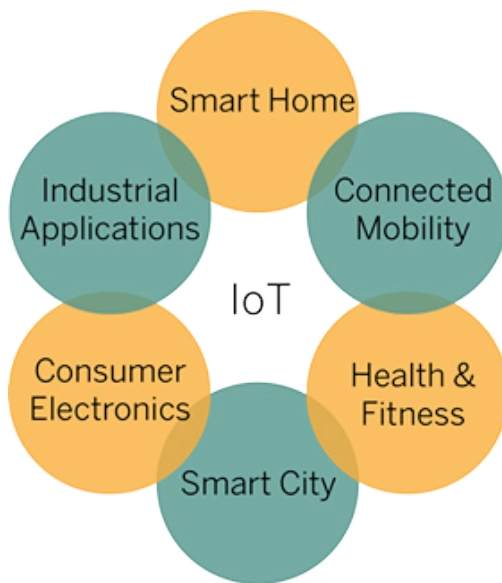


Figure 1. The Internet of Things connects several spheres of electronics.¹⁰

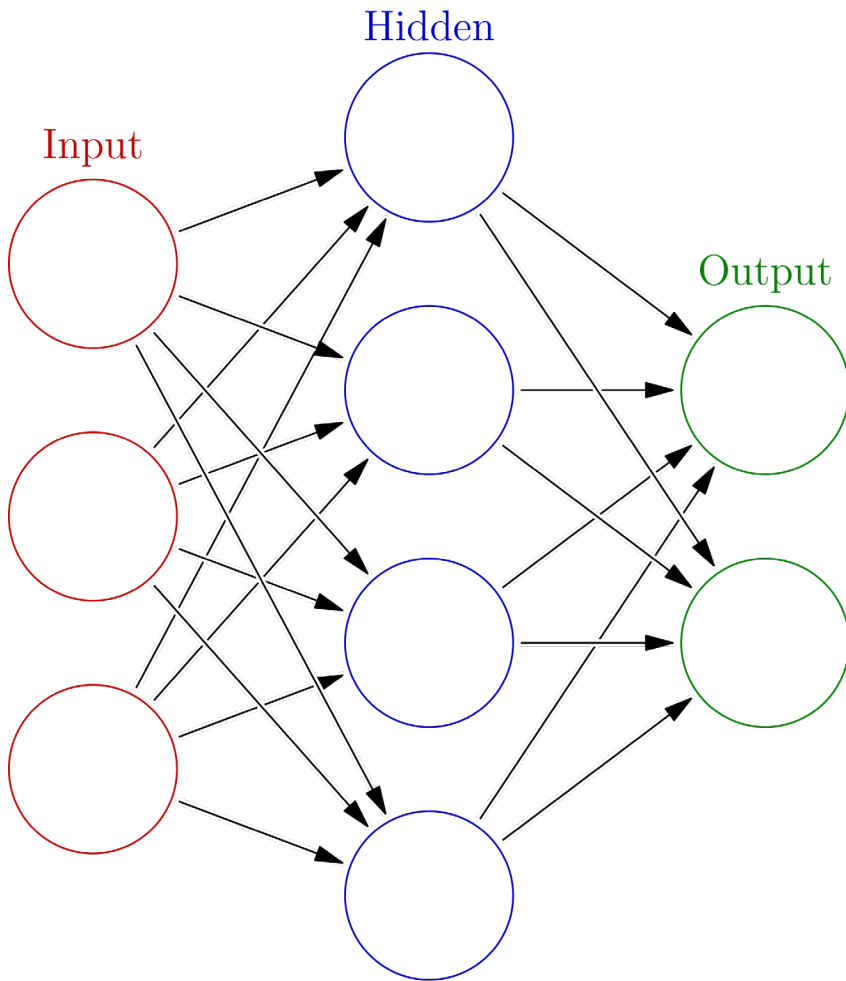


Figure 2. Artificial neural networks are commonly composed of three layers: input, hidden, and output. The input layer contains passive nodes, which don't modify data, while the hidden and output contain active nodes which do modify data.⁹

“By analyzing the details of past attacks on a network, deep learning can uncover attack patterns to fix security flaws.”

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vices, but also those of everyone else on the same IoT network. Machine learning can be used to recognize patterns in consumer behavior and predict their behavior in order to recognize those that show risky cyber hygiene, such as using manufacture default passwords and forwarding phishing links.³

The next generation will deal with security issues that extend beyond what information can be stolen from existing databases. Instead they must grapple with how that information can be used to produce new details about their private lives, as well as the smart devices that will dominate their lives and become a privacy liability. It is critical for consumers, even now, to be aware of potential security vulnerabilities in and data collection by the most basic internet-connected devices around them.

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TACKLING COMPLEX DISEASES WITH EPISTASIS

BY KARA JIA

Epistasis is a biological concept that was coined approximately a hundred years ago by the biologist William Bateson. Epistatic interactions, also known as gene-gene interactions, have been studied in relation to complex human diseases, such as Alzheimer's disease and multiple sclerosis, which are caused by a combination of genetic, environmental, and lifestyle factors. Some research proposes that epistasis further complicates the search for the genetic basis of such diseases; however, examining epistasis could be key to understanding complex diseases in depth.³ Recent research also presents new insight into the role of epistasis and its ability to help further scientists' understanding of several complex diseases.

Understanding the underlying cause of complex diseases has its basis in the two Mendelian concepts of inheritance: the principle of segregation (two members of a pair of alleles separate during gamete formation) and the principle of independent assortment (alleles assort independently from each other). Not all complex diseases exhibit simple patterns of Mendelian inheritance—that is, they do not demonstrate single-gene dominant or single-gene recessive Mendelian pattern of inheritance.⁹ Diseases such as cystic fibrosis and sickle-cell disease are caused by mutations in one gene, whereas more pervasive diseases, such as Type 2 diabetes and heart disease, are due to effects of multiple genes.

Bateson first used the term “epistatic” in 1909 to describe how a particular allele at one locus prevents the other allele from expressing

its effect.² For example, two loci, B & G, both influence hair color in mice.² Locus B has two possible alleles, B and b, and locus G has two possibilities, G and g. There are three possible phenotypes: black, white, and grey. Regardless of the genotype at locus B, any individual with copies of the G allele (genotype G/G or G/g) has grey hair. If the genotype at locus G is g/g, an individual with any copies of B allele has black hair since at locus B, B is dominant to b. Given that the genotype at locus G is not g/g, the effect at locus B is not observable since individuals with any copies of G allele have grey hair regardless of genotype at locus B. Thus, locus G is said to be epistatic to locus B.

Multiple definitions of epistasis have been given. Nonetheless, the presence of epistasis signifies that there is something of interest in the mechanisms and pathways involved in a particular disease (more specifically, the biological interaction between specific proteins) based on a qualitative assessment where the mechanism of one factor is affected by the presence or absence of another.² Epistatic interactions are also highly context-dependent; a disease-causing mutation in one individual does not necessarily have the same effect in all individuals.⁵ While detecting epistatic gene action may provide little value in revealing the underlying process behind a disease, the understanding of different modes of interaction between potential disease loci can help with the detection of genetic effects.²

Evidence for epistatic interactions comes from studies in model organisms such as



C. elegans (nematodes) and *Drosophila* (fruit flies), where systematic screens have revealed the presence of epistasis.¹ Since epistasis is so prevalent in these “simple” model organisms, the assumption is that they should occur in humans as well.¹ Experimentally, mapping epistatic interactions requires large sample sizes.⁸ Multiple hypotheses must be tested, by which a severe statistical penalty is easily incurred, and a large number of tests must be evaluated computationally.⁸ Because the effects of one locus (the target locus) vary depending on the allele frequency of an interacting locus, epistasis can have varying effects across populations.⁸ For instance, since an interacting locus’s allele frequency can vary among populations, the target locus’s effect could be significant in one population but not in the other.⁸ Many human diseases and disease-related phenotypes are quantitative traits whose variance is due to interactions between many different genetic loci.¹ Thus, the difference between biological epistasis (gene action) and statistical epistasis (variant alleles) should be noted.

Because more genes are expressed in the brain than in any other tissue, neuropsychiatric diseases are an ideal class of disease to study the role of epistasis in disease development.⁷ Multiple genes may be involved in a disease phenotype, and multiple gene-gene interactions may or may not increase disease susceptibility. However, a single gene, apolipoprotein E (APOE), represents multiple epistatic interactions for Alzheimer’s disease (AD), which is a complex neurodegenerative disorder that leads to memory loss and dementia.⁷ Researchers in the 1990s found that while carriers of one or more apolipoprotein E4 (APOE4) genes have higher risk of developing AD, not every individual who carried the genes developed the disease.⁴ This indicated the likely presence of oth-

er gene-gene interactions involved in the development of AD. Combarros et al. presented 27 significant epistatic interactions categorized into five groups: cholesterol metabolism, β -amyloid production, inflammation, oxidative stress, and other networks.⁴ These included synergistic interactions that yielded increased risk for developing Alzheimer’s, as well as antagonistic interactions that indicated a protective gene-gene interaction.⁴ APOE4 yielded the strongest interactions with three different genes: α (1)-antichymotrypsin, β -secretase, and butyrylcholinesterase K.

APOE variants have consistently been linked with clear evidence to late-onset AD (LOAD); many LOAD studies also demonstrate important epistatic interactions involving APOE.⁷ APOE’s effects are expressed through physiological processes such as cholesterol metabolism, which is diagnostically assessed by inflammation.⁷ However, it should be noted that APOE4 interactions are not necessary for predicting AD onset as the strength of APOE’s effects depends on a number of genes and pathways APOE interacts with to affect risk of developing LOAD.⁷

The immune system is also a rich source of epistatic interactions. Since epistasis operates at direct interfaces between proteins, understanding these interactions provides functional and genetic insight into the risk and susceptibility of acquiring autoimmune diseases. Immune cell function is controlled by receptor-mediated activation of intracellular signalling pathways, which initiate transcriptional and non-transcriptional processes affecting the cell state.⁶

“Bateson first used the term ‘epistatic’ in 1909 to describe how a particular allele at one locus prevents the other allele from expressing its effect.”

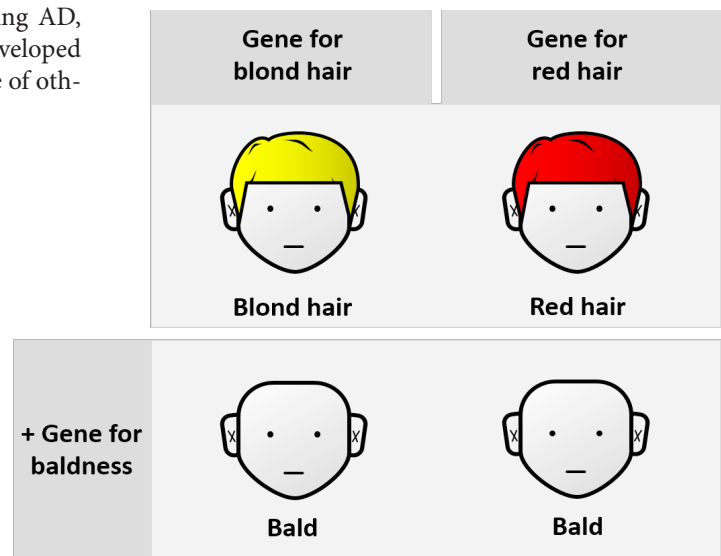


Figure 1. The gene for baldness is shown in this example to be epistatic to the gene for both blond hair and red hair.¹¹

“A single gene, apolipoprotein E (APOE), represents multiple epistatic interactions for Alzheimer’s disease (AD).”

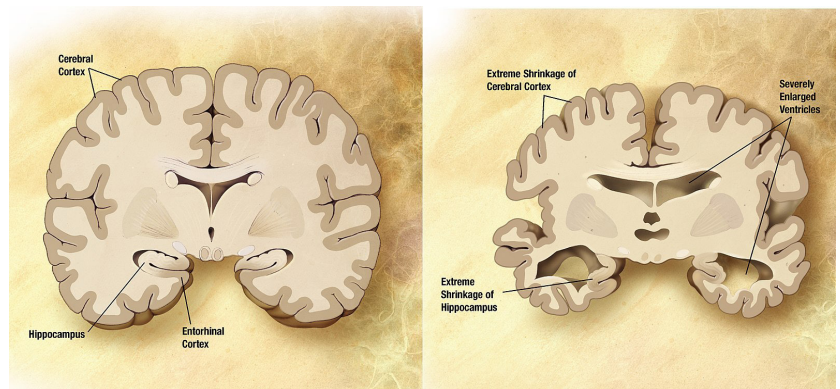


Figure 2. The APOE gene represents multiple epistatic interactions that are potentially responsible for developing Alzheimer’s disease.⁷

Epistasis occurs in multiple sclerosis, an inflammatory disease of the central nervous system in which activation of CD4+ T cells induces an influx of inflammatory cells that eventually causes demyelination, neuronal pathology and neurological dysfunction.⁶

Since treatments for various diseases are constantly being developed, understanding the genetics of diseases through epistasis could provide useful insight. Considering epistatic effects of a disease can potentially help provide better personalized disease treatments.⁷

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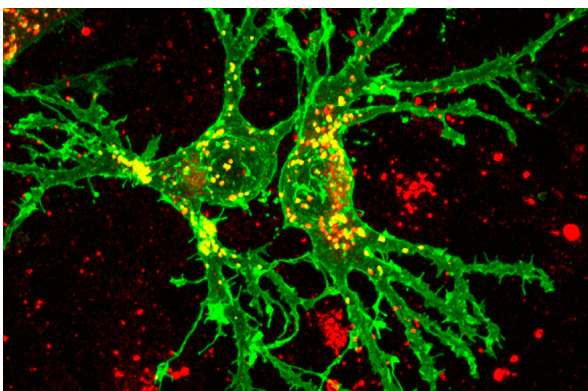


Figure 3. Epistatic interactions in multiple sclerosis induce demyelination, neuronal pathology, and neurological dysfunction.⁶



DRAWING THE LINE BETWEEN PREDATORS AND LIVESTOCK

BY LARA VOLSKI

In the blue, frigid night, a low howl sounds out over a cattle ranch situated in a vast rangeland of Montana. A guard dog lifts its ears in anticipation, and even the pine needles seem to prick with unease. At this point in the night, a livestock rancher has two alternatives: she can reach for her gun and try to put an end to the nearby wolf, or she can remain in her bed, content with the fact that she has taken preventative action to protect her herd. If she chooses the former option, she addresses the immediate threat but may risk disturbing the social dynamics of the local wolf pack and placing her herd in even greater danger. If she chooses the latter option, she aligns herself with the next generation of ranchers and herders, who utilize non-lethal methods to deter predators from preying on their livestock. Livestock management is an ancient practice that continues to serve as a foundational aspect to modern human society. Ranchers have handled conflicts with predators using the same method for centuries: by simply exterminating the problem animal. While lethal methods have proved to be fairly effective, there is a current movement

in the ranching community that emphasizes co-existence with top-of-the-food-chain animals. It starts with acknowledging that we play a similar role in the ecosystem as wolves, bears, or leopards, and will thus always be in competition with them for resources. Instead of wasting time pursuing problem animals after a calf or lamb has already been taken, ranchers are learning how to limit negative interactions before they occur. Predators are already learning how to adapt to us—African wild dogs, for example, have shifted to preying on small animals in lieu of larger ones (like livestock) when they share their territories with humans.² The next generation of ranchers believe that we humans can reciprocate this effort to adapt. Studies have shown that livestock depredation (loss of livestock to carnivores) decreases with abundance of natural prey and distance from protected areas.⁵ Humans can therefore differentiate the role of predator and rancher by maintaining wildlife parks and reserves and creating buffer zones between these protected areas and ranching operations. Yet the occasional depredation event will

persist as long as human encroachment continues, and it is a grave loss for any rancher to lose a calf. So why does the new ranching paradigm avoid the extermination of a problem animal when overlap occurs? Large carnivores are particularly sensitive to lethal methods because they invest more time and energy into parental care than many other animals and thus breed at a relatively slower rate.² We cannot afford to lose large predators because many of them are keystone species, which means their presence dictates how an ecosystem will function. Proponents of the new ranching paradigm even state that non-lethal methods may actually be more effective than lethal methods. The secret lies in working with the social communities of both livestock and predators. It can take a little imagination to place domestic cattle alongside their wilder, fiercer sisters that once ruled the vast plains of North America's heartland. Nevertheless, the instinctual wiring of the plains bison remains within the beef and dairy cows of today. These innate behaviors can be used to a rancher's advantage if she uses low-

“It starts with acknowledging that we play a similar role in the ecosystem as wolves, bears, or leopards, and will thus always be in competition with them for resources.”



stress herding methods to forge the herd into a self-protecting unit. Current ranching practices use fear and pursuit to move a herd from one place to another. A stressed cow will consequently want to return to the last place she felt safe, and abandon the herd in order to return to the previous pasture. Alone, she and her calf are vulnerable to depredation. But if a rancher focuses on low-stress herding techniques, a cow learns to see the herd as a place of safety. This will limit separation, strengthen social bonds, and reinforce herd mentality—ultimately encouraging the herd to defend themselves and each other from depredation in the same way that their wild bison relatives would.⁷ And the importance of social

bonds extends past the herd, too—guard dogs that bond with the herd as pups and are present for calf feedings are more effective at protecting the animals as adults.³ The next generation of ranchers also believe that non-lethal methods will help to promote robust social dynamics of large predators. Despite what one may guess, killing carnivores may actually lead to an uptick in livestock depredation. Think back to the rancher who has been disturbed from her sleep by a howling wolf. If she decides to shoot, poison, or trap this animal, she risks killing an alpha (the male or female breeding wolf). Without an alpha, packs have been shown to fracture into independent breeding pairs, who then all have pups of

their own and need to independently hunt to sustain each of their separate packs.⁹ Furthermore, if our sleep-deprived rancher ends up shooting a female breeding wolf, her mate may choose to adopt a polygamous mating strategy.¹ This would increase both the size and the appetite of the pack. It has been demonstrated, however, that these effects are brief and last no more than a year; some ranchers who rely on lethal methods are willing to risk a mere year in order to provide immediate protection to their herd.⁶ The next generation of ranchers believe that wolves are not the only predatory societies that are fractured by lethal methods. This antiquated ranching practice disturbs the bonds of another key predator: humans. A rancher is a guardian for both her herd and the pastures that nourish her cows, sheep, and goats. Fertile rangelands are among the most threatened ecosystems in the world because they are abundant in natural resources and economically profitable. With the threat of landscape conversion to residential and agricultural enterprises looming like a dark shadow over prairies and meadows, ranches have come to exist as some of the last asylums for threatened rangeland species. Ranchers are in this way a natural ally to preservationists, despite the fact that carnivore mitigation methods often drive the two parties apart. After all, rotational and timed grazing of cattle, sheep, or goats can be used to target invasive plant species and encourage the growth of native ones.^{4,8} Preservation can be of equal benefit to ranchers and preservationists by the practice of ensuring that wild prey populations are bountiful, which discourages the hungry bear or cougar from braving a field of cattle. By incorporating non-lethal and preventative methods into the next generation's ranching para-



Figure 1. Although we may not look similar, humans and coyotes play analogous roles in the ecosystem. We can decrease overlap by preserving wild game populations and by establishing buffer zones between protected areas like national parks and ranching operations.⁵

“Ranchers are in this way a natural ally to preservationists, despite the fact that carnivore mitigation methods often drive the two parties apart.”



Figure 2. The herd instincts running through these plains bison are also present within domestic cattle. By using low-stress herding tactics, a rancher can rekindle herd mentality within their cows, and liken them to their wild relatives.⁷

digm, we can draw a line between predators and livestock, and erase the line that exists between ranchers and preservationists.

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GOING PUBLIC WITH PERSONAL GENETICS

BY MARIE BALFOUR

IN THE MODERN DAY, PERSONALIZED GENETIC TESTS ARE INCREASINGLY COMMON.

Imagine walking into your doctor's office and having them pull up your entire biological profile online, tailoring your medical care plan to every molecular piece that makes you unique. This healthcare strategy is the future envisioned by the precision medicine (PM) movement, and it could potentially become the standard for patient care in a few years. As PM becomes more common, it has the ability to dramatically impact the healthcare industry through specialized healthcare solutions molded directly onto a patient's genome. Even though this approach represents the medical technology of the future, efforts should be made to understand and regulate disease dynamics models, adverse effects of testing results, and genetic discrimination that can arise from PM testing.

PM is part of the rise of the "personalized omics" or "multi-omics" movement in which multiple, comprehensive molecular testing mechanisms are used to assess individual profiles of biological molecules

such as RNA, peptides, fatty acids, carbohydrates, and gut microbiota. In 2015, the movement gained momentum through the implementation of the Precision Medicine Initiative by President Obama during his State of the Union address.¹ This approach seeks to integrate multiple lines of molecular data in order to get a wider view of the kinds of interactions which produce and maintain a disease in someone's body.² According to the National Institutes of Health, PM will take into account "individual variability in genes, environment, and lifestyle."³ One sub-section of this movement, genomics, has recently risen to popularity and is often used alone, without a full set of multi-omics data, to detect risk factors for certain diseases. Unfortunately, while genomic tests can help people understand their unique disease risk factors and plan for their future, they also open the door for genetic discrimination, oversimplification of the causes of disease, and unanticipated psychological distress. As this technology

begins to emerge at the forefront of medical advancement, it is important to understand and address the aforementioned challenges in order to properly assimilate precision medicine into modern healthcare.

Personalized genomics tests can help people understand their personal health profile, and this is one of the main reasons why personal genetic tests have been on the rise since their commercial inception in the 1990s and early 2000s through companies such as 23andMe, MyHeritage, and HomeDNA.^{4,5,6} In a recent study, one-third of participants surveyed on their reasons for undergoing genetic testing indicated that they were interested in having their genomes tested because of a known family history of disease. An additional quarter of participants sought to have their genomes tested in order to prepare for their healthcare future, even if they could not treat or prevent their genetic disease.⁷ These results suggest that the personalized genetics movement draws much of

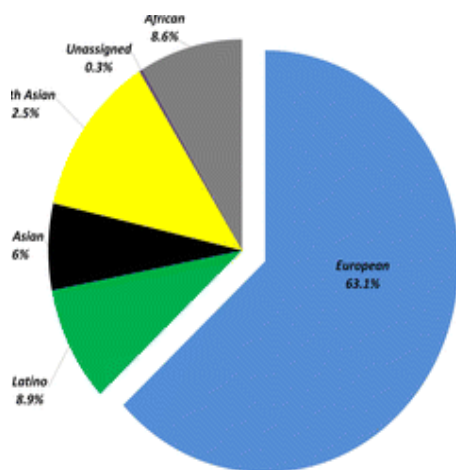


Figure 1. Percentage representation of ancestry/ethnic groups across 66,217 individuals tested by the Institute for Genomic Medicine and the Exome Aggregation Consortium in 2016.¹⁹

“In a recent study, one third of participants surveyed on their reasons for undergoing genetic testing indicated that they were interested in having their genomes tested because of a known family history of disease.”

its support from people who are genuinely curious about how their genetics can influence their lifestyle and healthcare plan.

As genetic testing increases in popularity, there has been a recent global effort to end discrimination based on genetic testing results, but legal action may not catch all of these genome-based injustices. The United Kingdom has already codified protections to prevent insurance premium increases for people who may receive unfavorable genetic testing results, but other countries, such as Canada, have struggled to pass bills that would protect those who participate in personalized molecular tests from discrimination.⁸ In 2009, the United States passed the Genetic Information Nondiscrimination Act (GINA) in order to

ensure legal protection against workplace discrimination based on genetics testing results. However, since GINA's implementation, at least one study has actually shown an increase in reports of genetic discrimination in the workplace.⁹ Because of this, some researchers argue that the act cannot address newly adopted testing technologies and the U.S. should add additional clauses to prevent further discrimination.¹⁰

Additionally, although personalized genetic tests are becoming increasingly common, the scientific community has not yet established a definitive causal relationship between genetic mutations and disease. Currently, there is not a plausible genetic interaction model for human disease based on personalized data that also

incorporates environmental and social factors such as diet, exercise, and socioeconomic status.¹¹ The multi-omics and PM perspective seeks to improve this outlook by gathering additional molecular data that provides a clearer picture of multiple markers for certain diseases. For example, microbiomics data analyzes human gut microbiomes, which strongly mirror diet patterns, an environmental factor. Results from the Integrated Personal Omics Profiling (iPOP) study, a longitudinal study assessing extensive biochemical profiling data, have suggested links between omics data and disease outcomes. Although iPOP results from microbiome, metabolome, and genome analysis have indicated a possible genetic association with disease indicators such as inflammatory response and mitochondrial dysfunction, more data must be collected in order to confirm these connections and integrate them into a conceivable disease model.^{12,13} Furthermore, there are still limitations to this method because not all environmental and social factors have known molecular markers that can be traced with multi-omics technologies.

Another complication to this testing approach is that the PM movement has the potential to misrepresent certain groups. According to recent analysis, patients of European ancestry are disproportionately represented in genetics testing, which makes it harder for scientists to analyze diseases in other populations.¹⁴ From these results, the precision medicine movement may have a Eurocentric bias that requires more testing from outside populations in order to paint a more accurate picture of many different types of human disease across the globe. Additionally, such mo-



Figure 2. President Barack Obama shakes hands with the Director of the National Institutes of Health, Dr. Francis Collins, after his remarks on the Precision Medicine Initiative on January 30, 2015.²⁰

Table 2 Motivations and concerns: themes emerging from the qualitative interviews

Themes
<i>Motivations</i>
1. To learn personal disease risk or health-related information (30)
To avoid or reduce their risk of disease (23)
Via lifestyle, diet and/or exercise (10)
Via medical intervention (4)
Prepare or plan for the future (9)
Emotionally prepare (2)
Financially plan (3)
Stored personal information could be of potential value to them in the future (4)
For medication prescribing purposes (ie, pharmacogenomics) (3)
For diagnostic purposes (1)
As the science advances (3)
Because of a known family history of disease (12)
Because of a known or suspected personal history of disease (5)
Because of a lack of information about family history (3)
2. Curiosity (23)
General curiosity (14)
Interested in or curious about the science (11)
The field of genetics/genomics (8)
Science (2)
Research (1)
Curiosity about their genes (7)
3. To contribute to research (15)
Altruistic desire to help others and the field of medicine (3)
To contribute to the advancement of science (5)
4. Interested in their ancestry (13)
5. To provide disease risk information for their children and other family members (11)
For their existing children (7)
For their family members generally (3)
For their future potential children and family planning (3)
6. Self-exploration (7)
7. Novel opportunity (3)
8. Useful professionally (1)

Figure 3. Motivations for genetic testing revealed by qualitative interviews from a 2016 study by Sanderson et al.²¹

“The process of acquiring informed consent from patients for these kinds of tests is complicated, because most patients may be truly unaware of what they are consenting to until they receive their results.”



lecular tests will most likely end up categorizing patients into set subgroups that match their molecular profiles instead of developing specific individualized medicines, which could have a stratifying effect on the healthcare industry that focuses more on the differences between genetic groups instead of the similarities that bind everyone together in the medical setting.¹⁵

Because genetic tests are just starting to become more user-friendly and commercially available, the “personalized omics” movement may not currently have the existing infrastructure to educate and support people who receive potentially distressing test results. Currently, personal genetics testing companies have protected themselves from liability for any discrimination or distress their results can cause, even

though reports indicate that many people who receive their testing results experience anxiety and psychological distress. Some researchers have even called for personalized genetics companies to provide genetic counseling services, in addition to their analysis software, to explain testing results to clients.¹⁶ The process of acquiring informed consent from patients for these kinds of tests is complicated, because most patients may be truly unaware of what they are consenting to until they receive their results.¹⁷ Even after patients have received their results, some may not be able to afford the medical therapies available for their personal disease profile.¹⁵ As these tests become increasingly popular and develop a larger pool of client data, there is also a concern for patient privacy. Although molecular testing companies promise total patient privacy and responsible data-sharing practices, unintended security breaches could have lasting consequences.¹⁸

In many ways, the scientific community has missed the opportunity to preemptively regulate the PM movement, and it is quite possible that some or all of its tenets will become commonplace in the next generation. PM technology provides a streamlined way for lots of people to receive their biological information at steadily decreasing costs, making this movement increasingly popular. However, although PM shows great promise, care should be taken to properly address disease dynamics, adverse effects of testing results, and genetic discrimination as the movement progresses.

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CROSSING THE SYNAPTIC CLEFT: TREATING AUTISM SPECTRUM DISORDER

BY SANIKA GANESH



VACCINATIONS

An infamous 1998 publication by Andrew Wakefield erroneously suggested that behavioral disorders, including autism, were linked to the MMR vaccine.¹ *The Lancet* retracted the article in 2005 after investigations revealed Wakefield's unethical methods, scientific dishonesty, and conflict of interests.² Since then, and as recently as 2014, scientists have been unable to find associations amongst vaccinations, autism spectrum disorder (ASD), and MMR (measles, mumps, and rubella).³ Yet parents are still apprehensive about vaccinating their children. While 94% of parents intend to vaccinate or have already vaccinated their children with the recommended vaccines, 77% of parents have concerns about vaccines, and 30% worry that "vaccines may cause learning disabilities, such as autism."⁴ The cause of autism spectrum disorder is much more complex than the anti-vaccine narrative proposes. Modern research aims to determine the exact cause of ASD and is headed in the direction of developing medical treatment for ASD.

According to the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-V), autism spectrum disorder is distinguished by impaired social interactions and communications, repetitive behaviors, and preoccupations with specific stimuli. Psychiatrists diagnose individuals with autism according to symptoms that fit these criteria, which are usually recognized by the third year of life. Because the disorder varies greatly amongst individuals, it exists as a spectrum.⁵

HERITABILITY AND GENETICS

In 1943, physician Leo Kanner first identified autism as a disorder distinguished by a lack of social skills.⁶ Kanner con-

"Hence, characterizing the relationship between what ASD-linked genes encode and how these genes are expressed phenotypically is the next challenge scientists face in determining how to medically treat ASD."

cluded that children with autism "come into the world with innate inability" to form normal relationships with other people because the children exhibited symptoms of autism at a very early age.⁷ Thus, by emphasizing the innateness of the condition, his research suggested the biomedical origin of autism.⁷ Yet, for the next twenty years, scientists hypothesized that bad parenting caused autism, claiming that children with apathetic and career-oriented mothers developed abnormally.^{6,2} Only in the 1980s, as researchers explored the heritability of ASD, did the biomedical explanation of autism prevail over the refrigerator mother theory.²

To determine whether autism spectrum disorder had a genetic cause, scientists examined concordance: the probability of two twins exhibiting the same physical or disease trait. Twin studies in 1977 revealed that autism spectrum disorder is highly heritable.⁸ Concordance rates of ASD are higher amongst identical twins than fraternal twins or siblings, indicating that the likelihood of developing ASD increases as more genetic

“This imbalance of the E/I ratio is thought to generate an excess of activity in the brain, which impedes normal information processing and ultimately results in cognitive impairments.”

material is shared with an individual diagnosed with ASD.⁹ But a single genetic mutation does not account for more than 1% of all ASD cases, suggesting that the genetic foundation for ASD is incredibly complex.⁶ For this reason, scientists discuss the likelihood of developing ASD by referring to genetic risk factors.

In genetics, an organism's genotype refers to the genes that it carries, while its phenotype refers to the observable effects of these genes. An individual may genotypically possess the rare deleterious mutations that confer high risk for ASD but not express ASD phenotypically. The process by which an individual's genetic background modulates the phenotypic consequences of deleterious genetic variations is called genetic buffering. Individuals with high genetic buffers are more likely to alleviate the effects of high-risk mutations and lower their risk of developing ASD. On the other hand, individuals with low genetic buffers can develop ASD from a high frequency of low-risk mutations.⁹

Researchers have made significant progress in determining what constitutes a high-risk mutation. One study has identified 107 high-risk genetic mutations that occur amongst 5% of autistic subjects.¹⁰ Both genetic factors and environmental factors account for the development of ASD, but genetic factors currently offer more potential for medical treatment. Hence, characterizing the relationship between what ASD-linked genes encode and how these genes are expressed phenotypically is the next challenge scientists face in determining how to medically treat ASD.

Figure 1. The MMR vaccine.¹⁷ Studies indicate that some parents are still hesitant to vaccinate their children.



SYNAPTIC PLASTICITY

Many genes associated with ASD affect synaptic plasticity, the ability of synapses to adapt to changes in activity. Mutations in these genes disrupt communication amongst neurons by altering the strength of inhibitory or excitatory synaptic inputs.⁹ The E/I ratio hypothesis specifically describes how this disturbance in neural networks occurs: imbalances in excitation (E) and inhibition (I).

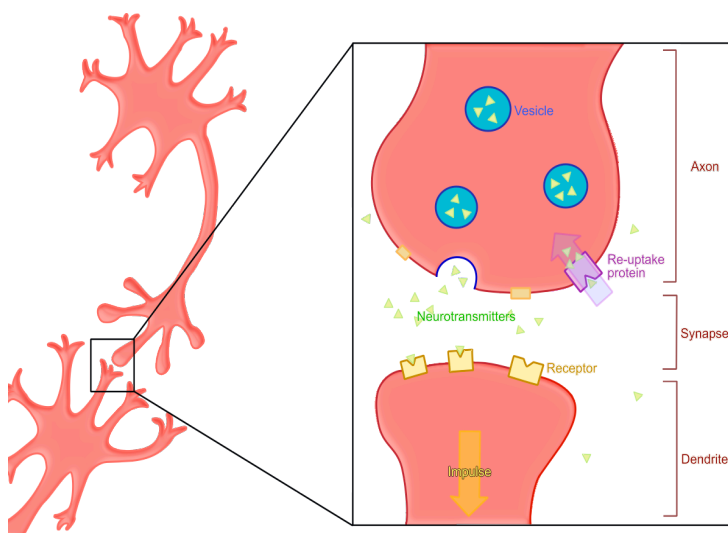


Figure 2. Synaptic function.¹⁸ Synapses regulate communication between neurons through chemicals called neurotransmitters. At the synapse, neurotransmitters from an axon terminal of a presynaptic neuron—the neuron sending the message—travel to dendrites of a postsynaptic neuron—the neuron receiving the message. The message received by the postsynaptic neuron, the synaptic input, can be excitatory or inhibitory. While excitatory synaptic inputs increase the likelihood of an action potential, inhibitory synaptic inputs decrease the likelihood of an action potential. An action potential occurs when a flow of ions moves across the axon of a nerve cell, permitting a neuron to communicate with the next neuron. Altering the behavior of synapses impacts how an organism processes external stimuli by influencing the likelihood of an action potential.

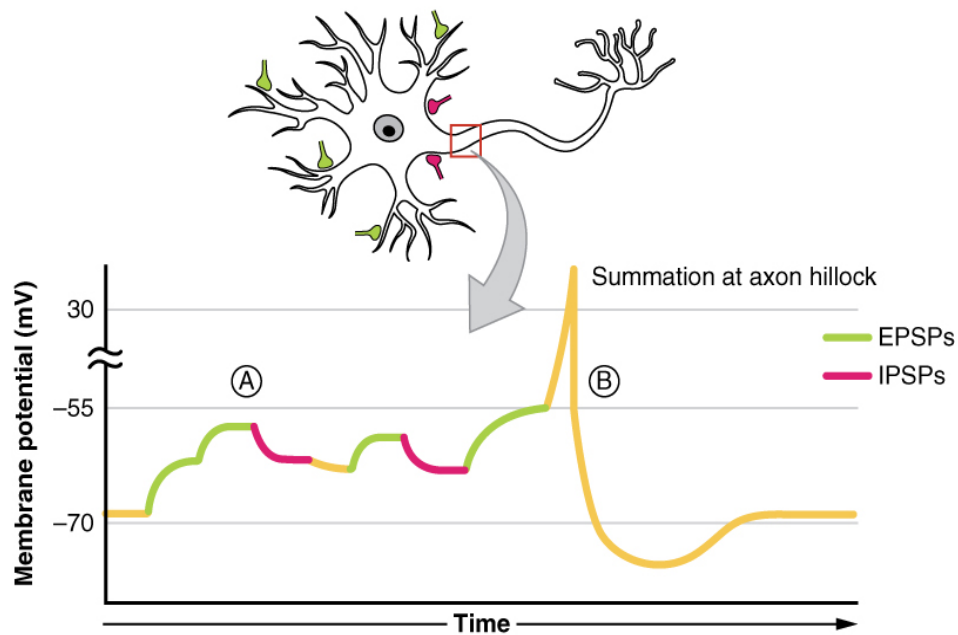


Figure 3. Excitation and inhibition.¹⁹ Excitation and inhibition ratios determine how excitatory postsynaptic potentials (EPSPs) and inhibitory postsynaptic potentials (IPSPs) influence the membrane potential of a neuron. The neuron fires an action potential if its membrane potential passes a critical threshold. Action potentials follow an all-or-none principle. Part A of the figure depicts EPSPs and IPSPs, while Part B portrays the action potential that ensues after an EPSP crosses the critical threshold.

What is the E/I ratio hypothesis? Many researchers predict that mutations in ASD-linked genes lead to fewer functional PV-interneurons.¹⁴ By inhibiting action potentials, PV-interneurons decrease local neuronal activity, expressing a protein called parvalbumin (PV) in order to regulate their firing rates.¹¹ Thus, if there are fewer functional PV-interneurons, excitatory cells receive less inhibitory synaptic neurotransmission, causing a decrease in inhibition relative to excitation.¹¹ This imbalance of the E/I ratio is thought to generate an excess of activity in the brain, which impedes normal information processing and ultimately results in cognitive impairments.^{12,13}

Neurons have homeostatic mechanisms that prevent too much or too little spiking activity. Synaptic homeostasis adjusts synaptic strength to stabilize firing rates, which can counter the effects of deleterious mutations.⁹ But for individuals with ASD, neither genetic buffering nor synaptic homeostasis is enough to offset the effect of the mutations.⁹

Scientists have tested potential therapeutic interventions based on the E/I ratio hypothesis. For example, drugs can alleviate at least some of the symptoms of ASD in mice by targeting the synapses of PV-interneurons to restore normal levels of inhibition.¹⁶ Likewise, optogenetic techniques (the use of light to control genetically modified cells) can increase spike rates of PV-interneurons in mice to compensate for the reduction in inhibition.¹⁵ Though these strategies have rescued some behavioral function in mouse models, neurobiologists must investigate whether the same can be done for humans.^{15,16}

From establishing the heritability of ASD to anticipating the implications of E/I imbalances, research regarding ASD

has made significant progress in the last fifty years. Though no medical treatment for ASD exists today, advances in implicated fields, such as genetics, psychology, and neurobiology, encourage scientists to explore potential options. It is possible that the next generation of researchers will successfully discover medical treatment for ASD in humans. In the meantime, the existing scientific literature can at least quell fears regarding vaccinations.

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THE FUTURE OF WORK

BY MACY CHANG

Self-driving cars are hitting the streets. Machines are databasing faster and more accurately than any office worker. Service robots are expanding from airport kiosks and grocery stores to serve restaurants, receive phone calls, build nanotechnology, and care for the elderly. As modern artificial intelligence reaches new heights in speech recognition, learning ability, and response to environmental stimuli, it is reasonable to wonder: which jobs will fall to the onslaught of robots? In the next couple of decades, the capabilities of AI are expected to overtake human performance and displace workers from nearly 50% of job categories.⁶ Along with forcing political change, this shift in the job market will impose an adjustment of our economic system as well as a reassessment of core American values, notably the value of work.

THE WINNERS AND THE LOSERS

The trend of unemployment as a cost of increasing industrialization is a precedent from the first three Industrial Revolutions. Respectively, the advents of engine-powered mechanics, mass production, and electronic information technology severed large parts of the labor force; however, as noted by many job market researchers such as Wayne F. Cascio, new types of jobs will provide new employment opportunities.¹

So far in our Fourth Industrial Revolution, which consists of rapid digital and physical technological integration, Cascio's prediction has proven correct; middle-class jobs have fallen and steadily been replaced by higher-skilled careers like programming and web design.¹ However, what sets this revolution apart is the unprecedented rate of machine adaptability. This leaves Amer-

icans to wonder whether human input will ever again be a baseline requirement for most tasks.

Whether or not a renaissance of new jobs is coming, short-term unemployment is an unavoidable consequence of high-level machine automation. The largest job losses are expected for a slew of middle-class jobs in the blue-collar and service sectors, particularly the transportation industry, personal vehicles industry, office and administrative jobs, sales, and construction.^{2,4} As described by researcher Carl Benedict Frey, these job types are within the 47% of jobs bound for automation in the next two decades. 19% more jobs are in moderate risk.⁴ Jobs involving creative and social intelligence, speaking or entertaining, and high-level scientific and technological skill are considered safe.¹

THE ECONOMY

The future of technological progress warrants a variety of ideas about how a human-machine hybrid economy would function. Regarding the outcome for the economy, there should be a distinct separation between the whole and the individual workers. On a grand scale, there are guaranteed benefits, including long-term GDP growth and significant returns for capital holders.⁸ The increasing efficiency and inexpensiveness of machine workers will lower capital costs for all major companies.

What is more undefined, however, is the fate of the workforce. Historical trends can justify how job loss will be mitigated by new industries. However, there is no guarantee this will occur; lack of need for human input may create "perpetual job anxiety", and even if some jobs were created, workers' rights might be discounted.^{6,8}



“The largest job losses are expected for a slew of middle-class jobs in the blue-collar and service sectors, particularly the transportation industry, personal vehicles industry, office and administrative jobs, sales, and construction.”

Other perspectives combine possibilities of capital gain with job creation, offering the idea that “ubiquitous technology” will benefit companies while being used to increase human efficiency rather than displace workers.¹ All of these perspectives are contingent on effectiveness and adaptivity of government policy.

POLITICAL COMPENSATION

Mass automation will force the American political system to conform through revisions of labor laws, corporate restrictions, and school curriculums. Secular stagnation and a diminished middle class are a reality with a “business-as-usual” approach, meaning that no government action is taken to mitigate unemployment.³ However, as economics researcher Young Joo Kim clarifies, government intervention is highly likely to occur before any major unemployment crisis. He considers that, even in a scenario where there are not enough jobs to go around, radical social programs with government-pro-

vided incentives may be a viable reality.⁷ Though incentivized social work is one viable future of employment, there are a great number of other possible political responses to consider, including set quotas for human workers, science and technology-based educational reform, and guaranteed basic income.

A universal basic income (UBI) scheme is an especially contentious, but possibly suitable, solution for job loss. Basic income would entail annual grants to every citizen of a baseline amount of money, likely around the \$10,000 range. Groundwork for UBI has been set in other countries, including by Canada’s Manitoba Project and Finland’s current universal welfare experiment. Universal income could prove synergistic; studies performed in North Carolina, Namibia, and India all demonstrated correlations of basic income to increased literacy and test scores in school.¹⁰ Higher educational performance will be vital for employment in a future with increasing demand for technical skill, meaning that UBI could possibly be an effective mitigator for job automation.

There are evident drawbacks to the universal basic income scenario, including budget impact, dismantling of other welfare infrastructure, and, of course, the fear of an idle population. Universal basic income studies in North America found a 13% decrease in work hours per family, mostly among mothers and teenagers.¹⁰ These studies and scenarios of UBI, despite entailing much controversy, are worth careful examination, as are other political mitigation techniques to handle potential job automation in the future.

THE VALUE OF WORK

As a final point of consideration, how will American social values manage the pressure of workforce automation? Digital technology will strengthen class opposition toward capital holders as the lower and middle classes face the short-term consequences. In the long run, increases in efficiency will likely be paired with increasing class inequality.⁹ From here, there are a few possible pathways. Human-machine interactions may become more natural over time and promote integration into the workplace.⁵ On the other hand, class opposition may be so contentious that political change is obliged to maintain social order. The middle route, and most adaptive route, is large-scale educational revision to create a science and technology-orient-



Figure 1. Sophia, a social humanoid robot created by Hanson Robotics Ltd., at the AI for GOOD Global Summit in Geneva, Switzerland. In the modern age, new heights in machine learning and visual data processing are being achieved.¹¹

“Mass automation will force the American political system to conform, including through revisions of labor laws, corporate restrictions, and school curriculums.”



Figure 2. A third generation Cruise Automation Chevrolet Volt driving on the streets of San Francisco. Self-driving cars are expected to cause major upsets in the transportation and personal vehicle industries in the next few decades.¹²

ed focus in school, training future workers to enter emerging industries.⁷ What amplifies this possibility is the estimate that twice as many students will be in higher education by 2025.⁶

A core principle of American culture is to be self-made—to work for what you have—and may stem from the Marxist idea that equates the input of labor to how much labor can be purchased.¹⁰ This value is intrinsic to the economy and all of American capitalist politics. However, as the U.S. sits at the epoch of a new Industrial Revolution, how will Americans change their laws and values when, like agricultural production, most of the work is done by machines? Whether future digital technology is used for workforce integration or corporate benefit, whether the government can create a comprehensive game plan, and whether Americans can adapt their values to a machine-hybrid society will all be powerful determinants of artificial intelligence’s impact in the next generation.

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POWERING THE FUTURE WITH HYDROGEN FUEL CELLS

BY MINA NAKATANI



“Similar to lighting a match, the ensuing chemical reaction releases usable energy; the difference is that instead of creating carbon dioxide as a byproduct, only water is produced, a much greener process.”

Back in 2010, the infamous BP oil spill in the Gulf of Mexico sent news stations scrambling for photos of the catastrophe. Images of the ocean, stained with a swirling mix of slick blacks and sickly browns like some sinister Impressionist painting, flooded the news. Before long, discussions circled around to questions of clean energy and the future of energy production. In recent years, one promising source of clean energy—the hydrogen fuel cell—has begun to show the potential to make significant changes to the face of industry, emerging as a major possible fuel source in the near future. Energy production at present comes largely from the burning of fossil fuels, such as gasoline. In much the same way as striking a match or lighting a fire, burning fossil fuels ultimately gives off energy in the form of light and heat. Unfortunately, the

byproduct is carbon dioxide, a greenhouse gas whose emissions have increased significantly since the turn of the century.⁸ Hydrogen fuel cells offer a promising alternative, storing hydrogen gas and providing a surface on which it can react with oxygen. The ensuing chemical reaction releases usable energy as well; however, because only water is produced as a byproduct rather than carbon dioxide, it is a much greener process.⁴ Nonetheless, as with any new technology, hydrogen fuel cells still encounter problems that do not have an immediate solution. One such problem is the physical storage of hydrogen itself. Compressing a gas in any scenario is dangerous, and attempts to do so in an unsafe manner can lead to rather explosive results. This problem is only compounded by the fact that hydrogen tends to erode its containers, even without external

factors—such as emptying and refilling tanks, or simply the chaos of everyday life.⁶ There are several methods being considered in order to combat this issue. One possibility involves the use of materials known as metal hydrides—positively charged metallic atoms bonded to negatively charged hydrogen atoms—to store hydrogen. Ideally, in this form, the hydrogen being stored is not hydrogen gas, but effectively dormant hydrogen ions, which do not undergo the rapid expansion, better known as an explosion, that hydrogen gas can.³ Although metal hydrides can undergo potentially dangerous reactions and retrieving hydrogen from this form is not easy, storing hydrogen in this manner nevertheless reduces risks by avoiding an explosive gaseous phase.³ More detrimental to the use of hydrogen fuel cells, however, is that the chemical

“The problem is fairly obvious; platinum is scarce and extremely expensive, making any large scale application too costly to be reasonable.”

processes which must occur are kinetically unfavorable—on their own, they are either slow or unlikely to occur at all. For the reaction between hydrogen and oxygen to occur at any reasonable rate, a precious metal catalyst, such as platinum, is required.¹ The problem with this is fairly obvious: because platinum is scarce and extremely expensive, any large scale application is too costly to be reasonable. Moreover, lacking knowledge of how exactly the reaction takes place, there is no definitive way to eliminate platinum from the fuel cell entirely.¹ Nonetheless, there are some alternatives, including alloys and the employment of different geometries. Research is being done on using metal alloys—mixtures of metals—made up of platinum, with more com-



Figure 1. Some fueling stations for hydrogen-powered vehicles do exist, but the infrastructure is still lacking for wider use.⁴

mon metals such as nickel or copper acting as a filler of sorts.⁵ The use of various geometries is also being researched: forming the platinum into thin sheets or nanowires, using inexpensive, non-precious metals for the core.⁶ In both cases, the same logic is used: exposing the greatest amount of platinum surface to catalyze the reaction, all while using the minimum amount of platinum overall. Though the ideal solution to this problem would come from understanding the reaction itself, these methods are a start. Beyond that, there is also the possibil-

ity of replacing platinum entirely, using replacement materials like molybdenum and tungsten, in conjunction with other metals, in order to replicate very similar catalytic properties.⁵ Combining that with research into the effectiveness of different geometries—thin sheets, nanowires, and possibly other two- or three-dimensional shapes—could render the problems surrounding platinum obsolete in the years to come.⁵ In addition, hydrogen fuel cells also have logistical problems to overcome. Much like the need for new charging stations to service electric cars, new infrastructure and equipment would be needed to refuel cars with hydrogen, all of which may prove costly to implement. For that matter, only 74 stations are predicted to exist in all of California by 2020.⁴ In order for hydrogen to be considered a major energy source, it would require much more infrastructure than that. Moreover, currently, due to the scientific and technological issues that remain to be solved, the price of a fuel cell vehicle is still higher than that of a comparable internal combustion—or even electric—vehicle.⁹ Even then, any future of hydrogen fuel cells hinges largely on public opinion; people often tend to place more trust in the known, feeling less safe with newer technologies, regardless of scientific potential.⁹ Yet the future of hydrogen fuel is promising, even if its coming may be slow. It already has begun to impact the automotive industry, with Toyota taking to selling hydrogen-powered cars with plans to sell 30,000 per year by 2020.⁷ This move could ulti-

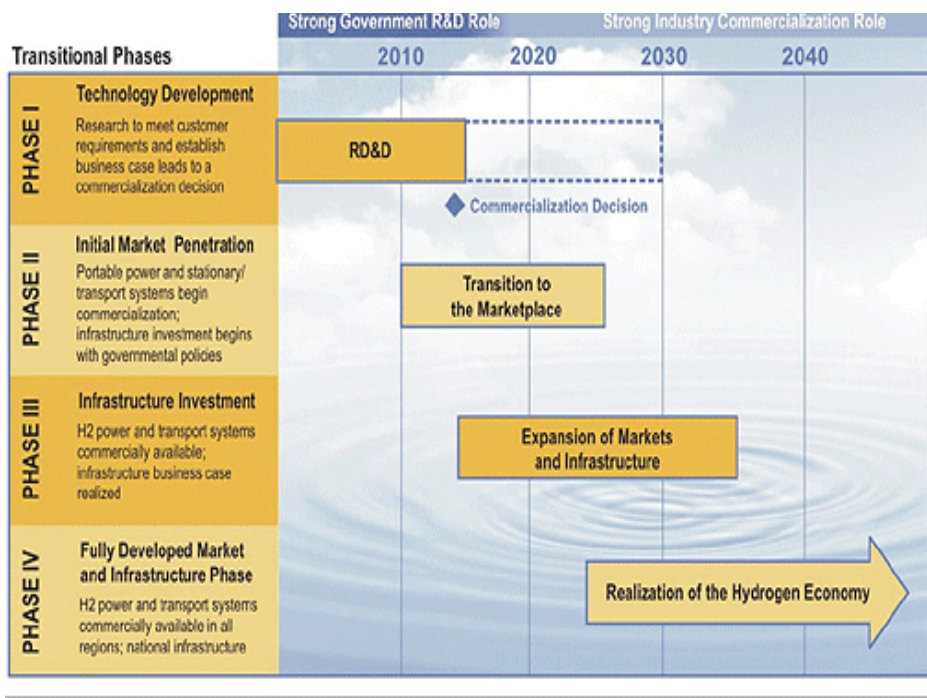


Figure 2. The planned development of hydrogen as a fuel source.

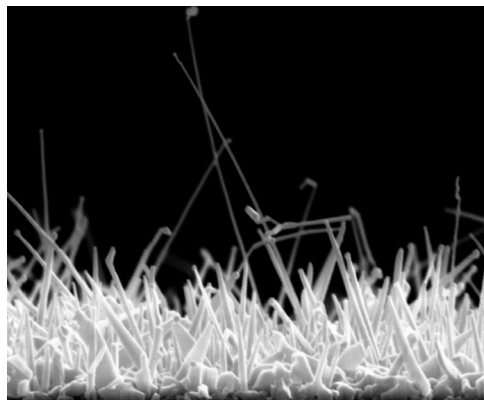


Figure 3. Microscopic view of metallic nanowires, a proposed geometry for platinum catalysts in hydrogen fuel cells.⁶

mately lessen the price of these vehicles and force the creation of more infrastructure, especially as the price of gasoline is likely to rise.⁹ In addition, the U.S. Department of Energy has plans in place to gradually implement hydrogen as a major fuel source, moving away from carbon-based fuels with the intention of improving technology such that 90% of energy will come from hydrogen by the year 2080.⁸ More intensive research would also allow hydrogen fuel cells to ultimately move outside solely the automotive industry. On a larger scale, it could be used for purposes like heating buildings;⁹ on a smaller scale, it can theoretically be used to power handheld diagnostic devices, such as pregnancy tests.² Overall, despite the problems still facing hydrogen fuel cells, their potential, even in diverse applications, is quite promising for the future of energy.

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NANOMEDICINE AND ITS VARIOUS APPLICATIONS

BY NICOLE XU

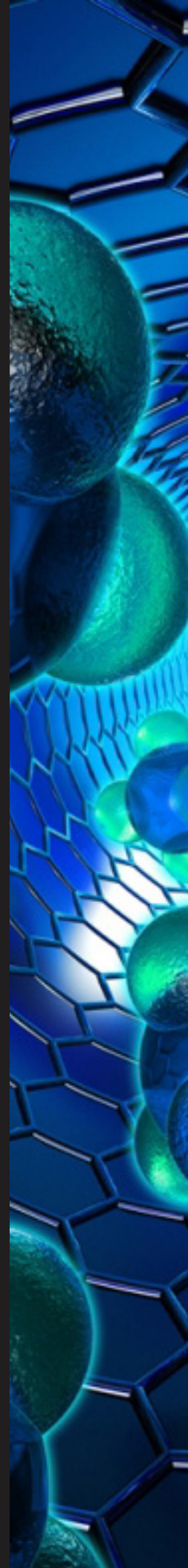
Walking into a drugstore, it is hard to imagine that the medicines filling the seemingly never-ending shelves represent only a small fraction of the possible remedies for existing diseases. In addition to those that are yet to be discovered, others, like nanomedicine, are invisible to the naked eye. Nanotechnology, the uniquely flexible manipulation and application of particles at the nanoscale, has recently been integrated into all aspects of healthcare, from diagnosis to treatment. As our lives become increasingly digitized, nanoparticles allow for a “smart” device within our bodies that is designed to give doctors more information than ever before on medical conditions, without the need for highly invasive procedures.

As with all living organisms, the human body is never in homeostasis, but rather constantly changing. While this ongoing flux proves advantageous for natural injury repair mechanisms, such as in the case of bone fractures, it is difficult for outside treatments to mimic the dynamic nature of the body. Many materials currently used for bone fracture repair are not biocompatible and can trigger immune rejection, a problem common to all foreign objects entering the human body.⁴ Additionally, doctors run into the challenge of finding materials that are conducive to movement. However, nanofibers have been synthesized that can mimic the natural extracellular matrix of the cell.⁴ These fibers facilitate the healing process by replicating inherent bone structure, helping new bones transition to become fully functional and supportive of full human body weight. Aside from simulating the natural structure of bones, magnetic nanoparticles can also respond to external stimuli while embedded inside an injured area, to enhance therapeutic effects. Bones require stimulation during the repair process to retain functional tissue. Usually very mobile limbs are

often restricted during healing by casts and other materials for an extended period of time, and nanotechnology provides stimulating signals without negatively affecting surrounding cells.⁴

Similar advances have been made in dentistry, where nanoparticles are becoming integrated into commonplace procedures. Metal ions have been widely used in treatments, and when combined with nanoparticles, there is an increased antibacterial effect.³ Together, they disrupt bacterial growth by interfering with signaling pathways or altering the bacterial microenvironment. Nanoparticles can also be integrated into dental fillers, periodically releasing minerals like calcium and fluoride to decrease chances of cavities and maintain dental health.³

Other than applications in treatment, nanomaterials can also be applied to the diagnostic process through imaging techniques. Gold nanoparticles can increase the contrast in CT scans without creating more toxicity for the patient.² Additionally, the adjustable properties of nanoparticles allow for chemical bonds, or conjugation, with other molecules, creating specific targets for imaging and differentiation between tissues.² The specificity of nanoparticles is beneficial for clinical applications as CT scans are often used as a diagnostic tool, and a slightly unclear image could cause a doctor to overlook a potentially threatening disease. Early detection increases the chances of successful treatment, so this serves a very practical purpose. Also, CT scans are often used prior to surgeries to map out the path of entry, so the more distinguishable that different tissue and organ structures can be, the fewer surprises the surgery team will face once in the actual operating room. Imaging techniques take advantage of the “smart” qualities of nanoparticles, using these particles to collect real-time data



about various pathways or target areas of the body. Nanoparticle probes are used with MRIs as they are designed to respond to a certain pH or temperature, which can then pinpoint tumors in contrast to normal healthy tissue.² Additionally, these probes can also reflect the internal conditions of identified tumors.

Even after diagnosis aided by nanotechnology, doctors still face the challenge of successful treatment. However, many common diseases currently lack cures or effective treatments. The human body contains many natural barriers against outside harm. These barriers are difficult to breach with traditional medical techniques. The blood-brain barrier, for example, is a semi-permeable membrane that protects the brain from infection while allowing key nutrients to flow through. Many debilitating diseases hide behind this barrier, such as Alzheimer's disease and Parkinson's disease. Scientific consensus is that the accumulation of amyloid-beta plaque between nerve cells partially causes Alzheimer's. Nanogels, which are nanoparticles composed of crosslinked polymers, provide an effective way to prevent the aggregation of the plaque in nerve cells, thus lessening the effects of Alzheimer's.¹ Nanoparticles can also provide the growth factors needed in Parkinson's disease in order to increase dopamine levels and promote overall brain development. Similarly, in multiple sclerosis, the myelin sheath surrounding nerve cells is damaged, but myelin-coated nanoparticles can act as a replacement while the immune system is degrading the naturally occurring myelin. Another barrier present in the body is the blood-labyrinth barrier at the inner ear, which refers to the different chemical compositions of inner ear fluid and blood.¹ In many patients suffering from inner ear disorders, surgery is needed to repair the cochlea, a cavity in the inner ear that converts sound vibrations to neural impulses, but nanotechnology can help bypass surgery and help patients with partial hearing loss.¹

While these are all promising discoveries, nanotechnology must become more cost-effective before becoming widespread. It is already difficult to provide affordable healthcare for a growing population, and the high cost of nanotechnology only adds to the existing challenges. Many nanotechnology-based medicines are currently marketed by small startups and businesses, and it is difficult for these companies to branch into a market dominated by large pharmaceuticals.⁵ Once these obstacles are overcome, nanotechnology and its applications in the clinical field hold great promise for more effective diagnosis and treatment strategies.

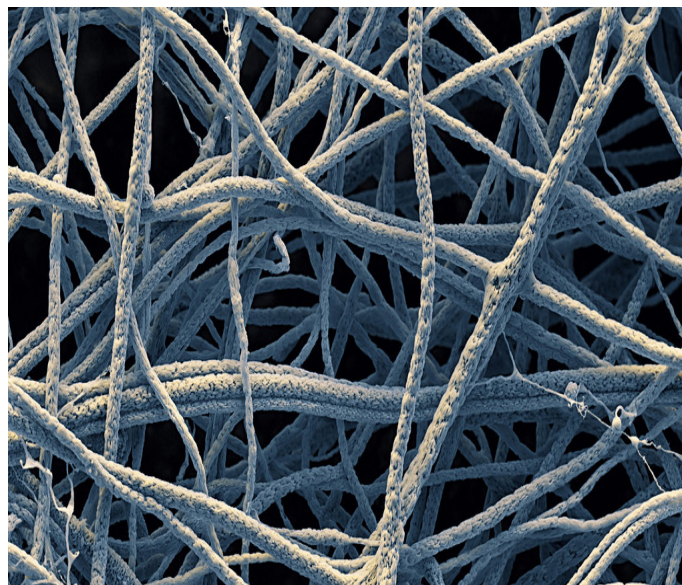


Figure 1. Inorganic nanofibers. Artificial nanofiber matrices can display a similar structure as individual fibers and networks in human bone extracellular matrices, helping to facilitate bone repair during fractures.⁴

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Shifting Power Dynamics: The #MeToo Movement

INTERVIEW WITH PROFESSOR DACHER KELTNER



Professor Dacher Keltner

BY ARJUN CHANDRAN, CASSIDY HARDIN,
MICHELLE LEE, MELANIE RUSSO, & YANA PETRI

Dacher Keltner is a Professor of Psychology at the University of California, Berkeley, where he directs the Berkeley Social Interaction Lab. He is also the founder and director of the Greater Good Science Center. Professor Keltner's research interests include emotions, social interactions, power, and behavior. In 2016, he wrote the best-selling book *The Power Paradox: How We Gain and Lose Influence*. He also served as a consultant for the Pixar movie *Inside Out*. In this interview, we discuss the shifting power dynamics in modern society, the influence of power on individual behavior, and the effects of power inequality that lead to the #MeToo movement.

BSJ: How did you get into the field of psychology? How did your childhood influence your interest in studying power, social perceptions, and behavior?

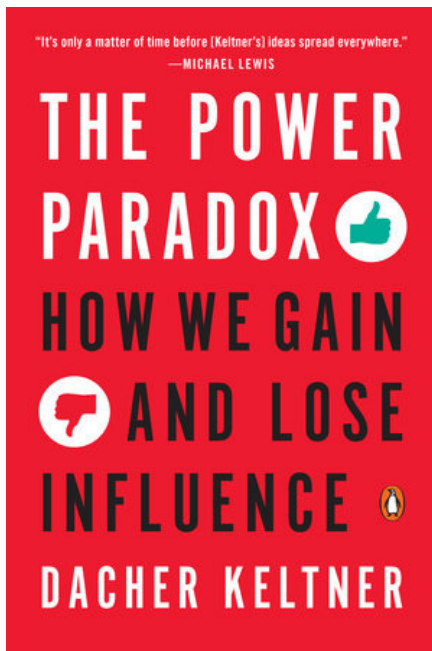
DK: I grew up in LA while my mom was getting her Ph.D. in the late 1960s. Then we moved out into the country in the foothills of the Sierra by Auburn. I think my parents are what got me into psychology. My dad is an artist, and my mom is a literature professor. In a way, art and literature are our deepest statements about psychology. It is hard to capture consciousness like literature does (I don't think science will ever get that close), or emotion like art does. I was raised in a special environment, going to museums, having mom read quotes from William Blake and Virginia Woolf. When I went to UC Santa Barbara for undergraduate studies, I felt inclined to study psychology as a discipline. But the big moment was my postdoctoral work with Paul Ekman. Paul, 84 now, pioneered the measurement of facial muscle movements. It took him seven years to figure out how to look at a face and know what facial muscles, which we share largely with chimpanzees, are moving. While in graduate school, I went to one of Ekman's talks, where he described how you could measure emotions anatomically. I had looked at paintings all my life, and I was just blown away by that idea. I got to be a postdoc with Ekman and that got me into the science of emotion. I study hierarchy and emotion, and hierarchy really means power, status, class, and inequality. I just wrote this book *The*

Power Paradox: How We Gain and Lose Influence, which summarizes everything that I had done for 20 years. I realize now that I became interested in poverty and power after my parents moved my brother and me from a middle class community in LA to an extremely poor rural town.

BSJ: You have developed a theory that addresses the individual and social implications of power. How do you define power?

DK: The social-psychological study of power is dominated by Cameron Anderson in Berkeley Haas, and Serena Chen, my colleague. We started that work 25 years ago, which led to our current theories. Economists think of power as money, political scientists think of it as the right to vote. Yet, there are a lot of examples in history that have nothing to do with money and nothing to do with the political state. Berkeley is an inspiring example with the free speech movement, which led to the antiwar protests. These poor undergraduates who had no support faced military opposition and changed the world. I was not at peace with the idea that power was money, or politics, or military. We defined power as your capacity to alter the state of another person, or other people. This definition becomes really psychological. A lot of diseases during pregnancy are about the power struggle between the fetus and the mom. Younger siblings are the revolutionaries and older siblings are the fascists.

Figure 1. Keltner's book *The Power Paradox*, which summarizes 20 years of his research on power.²



To say there isn't power in these dynamics is ridiculous. Power does not really correlate (only by a factor of 0.2) to how much wealth you have. You can be a billionaire and not do anything for the world, or you can be a very poor person who changes something and does a lot for the world. Bertrand Russell, the great philosopher, says, "power is the basic medium where all social interactions take place," and I believe that. Love, sibling dynamics, mother and child, and work are all about power.

BSJ: In your Google talk, you mentioned that the Machiavellian way to power is now less effective than the empathetic way to power. Could you explain the difference between these two?

DK: Western scholarship pits two hypotheses against each other. One is Aristotelian. Writing in classical Greek times, Aristotle thought that power could be found through virtue. He argued that power took courage, kindness, impartiality, and empathy. The other is Machiavellian. Machiavelli was a politician who got kicked out of his job. He went to his country estate and thought about how he could get back into the political game. Finally, he chose to write a book about power to the Medici in order to get his job back. He wrote *The Prince*. This book is about lying, deception, leading through fear, ensuring your allies are weak. *The Prince* was fitting in one of the most violent periods of history and promoted weakening others for gain of power. About 1500 years ago, we were really hierarchical, male-centered, patriarchal... Slavery was okay, polygamy was okay, and expending human life for your own gain

was acceptable. Slowly, the civil rights movements have moved us away from these things. There's a lot of data that shows that, pre-Trump, we were moving toward the more empathetic and Aristotelian form of power, where people would sacrifice for others and build social networks. Now, sadly, we have the rise of the strong men again.

BSJ: Before we talk about how power influences behavior, we would like to ask you some neurophysiology questions. First of all, what neurological changes are brought about by power?

DK: Our behavioral data indicates that, when you are in power, you lose the ability to take the perspective of

"...pre-Trump, we were moving toward the more empathetic and Aristotelian form of power... Now, sadly, we have the rise of the strong men again."

others. For example, we have seen that the more powerful person in a relationship often has trouble reading their partner's emotion from a photo of the face, so there is what we call an empathy deficit. One of my students studied the vagus nerve, which is the biggest bundle of nerves in the human nervous system. We found that the vagus nerve tracks compassion. My student interviewed privileged people with power and showed them images of suffering: kids with cancer, kids in a famine... People with less power had a vagus nerve response, and people with a lot of power did not have a vagus nerve response. Overall, we found that the empathy-compassion network gets deactivated by a certain amount of power, with exceptions of course.

BSJ: Why do you think there are more men in power now even though physical dominance shouldn't be as important as intelligence?

DK: I think that gender inequality is the most important current issue besides climate change and mass incarceration. #MeToo is real, women are slowly rising in power compared to 60 years ago. The proportion of woman-senators is increasing, women are rising in the STEM fields... But the U.S. is a little behind other countries in terms of these trends. I was recently in Davos and I talked about sexual harassment. Harvey Weinstein is the perfect embodiment of the power paradox. He started a film company, made good films, became a big influence, and then turned into an animal—his social and financial power led him to unethical behavior. But that's all over Hollywood, Washington D.C., all over tech, and we know that now.

Compared to Canada, Finland, Iceland, which have a gender balance principle in their cabinet, the U.S. lags behind. Donald Trump is a deep throwback. Most people who support him have trouble with women in power.

BSJ: In the recent USA Gymnastics sex abuse scandal, more than 265 women accused Larry Nassar of sexual assault. What can we do in the future as a society to help women report earlier and ensure that sexual abuse is much less frequent?

DK: I would like to first direct you to an article: “Sex, Power, and the Systems that Enable Men Like Harvey Weinstein.” About 80% of women are facing sexual harassment at work, but it’s difficult to have accurate estimates because so many of them are silent. Until recently, we didn’t have a language to express this type of abuse. Women said, “Well, he did kiss me in the copy room, but it was a mistake.” We should re-think that. When people say that “Harvey Weinstein is a sex addict and he needs therapy,” I say bull. That is not the story. He is violent and he assaulted people. He does need therapy and he can pay for that, but the deeper and more troubling story is that anywhere—in Hollywood, Washington D.C., the tech industry, in universities—we have a social organization where the majority of those in higher power are male. Young women are coming in as assistants in Hollywood and 94% of directors are male. You can see that there is a system in place that makes it radically more likely for incidents like sexual harassment to happen. So, what’s the solution? Frank language and reporting with details. “This is what you did, and this is the policy. You cannot grab somebody’s ass.” Frankness and science-based precision. While teaching a group of leaders in Kaiser Permanente, I noticed that three out of the six medical directors of the company are women. It’s the first time in the history of medicine where there are this many women in such high positions. These women were inspired by *The No Asshole Rule*, written by Bob Sutton in Stanford. They created a set of policies to prevent sexual harassment. These policies explicitly stated what the employers could and could not do. With these policies, even in an alpha male environment, these women were able to speak and stand up for themselves. This example shows how important it is to establish the right policies to prevent sexual harassment. The second important thing is giving women power. If there was a board that makes executive decisions where someone is acting like Harvey Weinstein and if there were women on that board, there would be a difference in the decisions the board makes. It really is fundamentally about promoting equality and giving women power. Also, if there were just as many male interns as female interns, the situation in Hollywood would change. As you have more gender equality in work places, the issues regarding sexual harassment, assault, and rape diminish because women are there to hear the story, develop policies, and punish people accordingly. It is a power struggle.



Figure 2. Keltner served as a consultant for the movie *Inside Out*, which suggested that allowing yourself to feel every emotion—including sadness—is a fundamental part of life.

BSJ: What do you think is the psychological reason behind the victims’ silence?

DK: There are many studies that inquire how power changes the brain and how we think and behave. It’s called the power of approach and inhibition theory. People in power often are able to say what they want and take what they desire. However, people who do not have power feel constrained in various aspects. People in lower power feel a greater threat, have higher cortisol levels, greater sympathetic autonomic ner-

“#MeToo is real, women are slowly rising in power compared to 60 years ago... but the U.S. is lagging behind.”

vous system activation, and greater blood pressure. Their minds are more inhibited; when they speak, they hesitate, interrupt themselves, and [do] not say what they actually think. Imagine being a young woman of lower power, facing a committee full of males. Silence becomes an obvious ending to that story.

BSJ: Moving away from gender dynamics and focusing more on the hierarchy of power, how, in your opinion, could we prevent people in higher power from behaving unethically? Could you possibly relate this to Trump’s administration?

DK: People in higher power are more likely to cheat in a game where they can win money, take candy from children, and lie. You can tell how power makes individuals

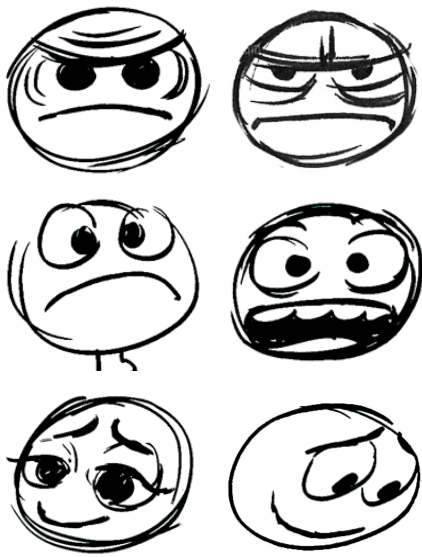


Figure 3. Emojis from the “Finch” Facebook sticker pack that Keltner and artist Matt Jones designed together. Emotions of anger, terror, and maternal love (two representations in each row). Image courtesy of Dacher Keltner and Matt Jones.

prone to unethical behavior from the study I conducted in Berkeley regarding how owners of different cars would drive past pedestrian zones.¹ About 46% of those who owned more expensive cars would drive past the pedestrian zone without stopping, while those who owned poorer cars would be more likely to stop every time at a pedestrian zone. Such examples are everywhere. Officials of the administration, like the Veterans Affairs Chief, might take their wife on a trip to Paris with a budget created by taxpayers’ money and justify it as a necessary act. People in power become blind to the unethical decisions they make because they justify it as a risk they needed to take as a person in power. This becomes relevant to the Trump administration when social psychology and social networking is involved. When someone of higher power is exposed on a social network and when people are able to comment on their behavior, leaders become inclined to act more ethically. This is a foundational trait that holds people accountable for their actions, and is a role of journalism. Many leaders of nations, dictators or presidents, have been afraid of social critics. Trump goes after news, and Hitler and Stalin were obsessed with artists, because artists are also social critics. The Soviet art was transformed into propaganda during Stalin’s reign. Consequently, with respect to Trump, you need critical commentary and transparency, which is something that already exists in the science community through peer-review and open data.

BSJ: Could you tell us about your experience as a consultant for the film *Inside Out*? What’s the central wisdom of that movie?

DK: *Inside Out* has very good Berkeley connections. There are often Berkeley streets in Pixar movies—*The Incredibles*; *Monsters, Inc.* I have known director Pete Doctor for a long time. He asked me about expression and emotion. He was having a typical experience with his daughter, who is eleven and is starting to become distant and move into a pre-adolescence space. He called me up and said that he could make a movie about the turmoil of becoming an adult. Pixar likes to bring in psychology experts and ask them to talk about science. I went in every six months for two and a half years. A group of people works on the script, the story, and comes up with drawings. I met with these small groups and asked questions. How many emotions do the characters express? What are these emotions? How do children behave when they have these emotions? How do we accurately recall our childhood? There are two important things about *Inside Out*. First, all emotions are there for a reason. For example, fear helps you avoid danger. Emotions are not crazy out-of-control purposes—they are functional. Second, it’s okay to be sad. Pete once asked me: Is sadness different from depression? I told him that there is a big difference. You can be sad for a couple of years if you lose someone, while depression is almost a lack of emotions—an apathy. I didn’t know it at the time, but Pixar was actually at a crisis in the movie. Pete wanted Joy—one of the main characters—to go on a journey with Sadness. The executive team disagreed—they insisted that it should really be Fear. But Pete kept persisting and sadness turned out to be the right emotion to use! It’s amazing that after the movie people started asking me, “Is the central message of the movie that it’s okay to be sad?” Usually middle-aged men who are also going through a middle-age crisis ask me that. And I’m always like: “To-o-om, it’s okay to be sad!” And the middle-aged man starts crying and says that he watched the movie with his daughter and that it changed his ideas about life. Unfortunately, culture pushes us to think the opposite, that sadness is not okay. As a parent, I say to my daughter, “Okay, let’s do some push-ups! Let’s go on a hike! Here is some medication!” But it’s important for people to embrace their sad parts of life to achieve closure—it’s fundamental. When you have kids, remember to give them space and understand that sadness is a part of life!

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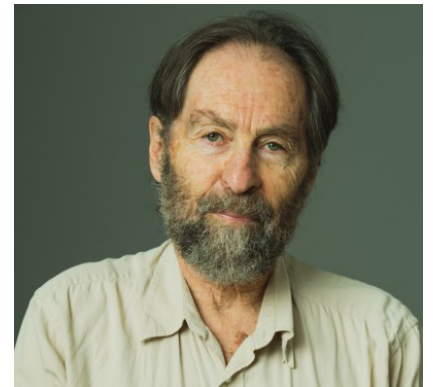


EXPLAINING PATTERNS IN ECOLOGY: CLIMATE MANIPULATION AND MATHEMATICAL MODELING

Interview with Professor John Harte

BY NIKHIL CHARI, ROSA LEE, PHILLIP DE LORIMIER, MOE MIJJUM, SONA TRIKA,
& ELENA SLOBODYANYUK

Dr. John Harte is a Professor of the Graduate School in the Department of Environmental Science, Policy, and Management and the Energy and Resources Group at the University of California, Berkeley. Professor Harte's research interests include ecological field research, the theory of complex systems, and policy analysis. In this interview, we discuss his investigation of climate-ecosystem feedback dynamics and his application of information theory to ecological systems.



Professor John Harte

BSJ: We read that your background is in theoretical physics. How did you transition to the field of Environmental Science, Policy and Management?

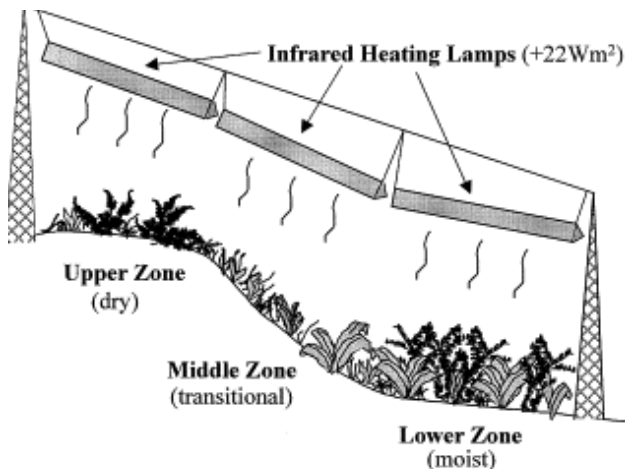
JH: After my postdoctoral work, I was appointed an assistant professor of physics at Yale. During my appointment, I discovered that I could make contributions to the fields of ecology and environmental science. The major event that caused this realization was my participation in a 1969 study in the Everglades. A plan was afoot to drain the Big Cypress Swamp and build a massive airport for supersonic passenger planes. We knew that building the airport would have a destructive impact on wildlife, but we were looking for other kinds of harm it would cause because the wildlife aspect alone wasn't going to stop the project. A colleague and I got interested in the subterranean hydrology of South Florida, and

we realized that draining the swamps for the airport would cause a massive amount of salt intrusion into the water supplies of people living along the Gulf Coast. We got some maps of the geology of South Florida, did several back-of-the-envelope calculations, and were able to show that if they built that airport there, half a million people would lose their freshwater supply. That argument convinced the Secretary of Transportation under President Nixon that it would be political suicide to destroy the water supply for Florida—a swing state. So, they cancelled the plans for the airport. And thus we used a little bit of physics to have a big impact on policy. That was a real watershed event in my career because it convinced me that you could do good science and actually influence policy. And so, I decided to switch fields and train myself in all the different areas of environmental science.

BSJ: What are positive and negative feedbacks in ecology, and what effect do these feedbacks have on climate change?

JH: Back in the 1980s, a group of scientists was doing a very exciting study. They were looking at a 2 km-deep ice core in the Antarctic, at a place called Vostok. When you look at that core, you're going back in time: each year's ice deposit is distinguishable. In any given year, you can extract two pieces of information. First, the bubbles of trapped air tell you what the atmosphere looked like when that ice layer formed. Carbon dioxide levels tended to be very low during ice ages and higher during the interglacial periods. The second piece of information is a global averaged temperature inferred by looking at the isotopes of oxygen in the ice. The ratio of heavy oxygen (^{18}O), which is not very abundant, to the common oxygen (^{16}O) tells us about temperature. Using these data, you see that temperature and carbon dioxide levels move in synchrony with one another. It gets more complicated, but the correlation is remarkable. Now, what does it mean? It means that when it's warmer, more carbon dioxide builds up in the atmosphere, which creates more warmth. That's a positive feedback. Cooling, in turn, pulls carbon dioxide out of the atmosphere, which causes more cooling, which pulls out more carbon dioxide. Such feedbacks in the Earth's climate system can be ocean-mediated, soil- and vegetation-mediated, or, most likely, both.

BSJ: We read about your experimental manipulations and long-term observational studies of an ecosystem in the Colorado Rocky Mountains. Could you tell us about the set-up of the climate manipulation experiment?



JH: In the late 1980s, I decided to experimentally heat an ecosystem that has a lot of carbon in its soil to try to understand feedback mechanisms. If we heat it, will the microbes speed up their activity and release carbon dioxide? That would be a positive feedback, and the heating would be exacerbated. Or maybe warming causes more photosynthesis and the plants take more carbon dioxide out of the air and put more carbon into the soil. That would be a negative feedback to global warming. We didn't know what would happen. We chose a large area of a subalpine meadow at the Rocky Mountain Biological Laboratory, where I had been working on other things for the previous decade. The experiment involves ten 10x3 meter plots on the side of a mountain. We built four tall steel towers and strung a web of cable from which we suspended electric heaters. We turned them on at the end of 1990, and they've been on ever since, running day and night, summer and winter—gently heating the ecosystem. The original idea was to see if we would discover mechanisms that on a larger scale would result in significant feedback to climate warming. Which, in fact, we did.

BSJ: One of the metrics you focused on in the warming experiment was loss or gain of soil organic carbon in response to temperature fluctuations. Why is this metric so important when discussing ramifications of climate change?

JH: Soil is a huge store of carbon. There's about five times more carbon in Earth's soil than there is in the atmosphere in the form of carbon dioxide. Potentially, this could be a big source of feedback. In our experiment, we mimicked the projected climate for the year 2050 assuming we keep burning fossil fuels. The experimental heating caused snow to melt a few weeks earlier each year, resulting in a longer growing season. In addition, in the first decade of the experiment, the plots lost 25% of their carbon—it was a dramatic change, and we were totally surprised. We started looking for a mechanism. There were some other changes that occurred simultaneously. Every year, we would see fewer and fewer wildflowers in the heated plots. A shrub—sagebrush—was replacing the wildflowers. Critically, the wildflowers are much more productive than the sagebrush; they pump a lot more

Figure 1. Schematic of a heated plot used in the Rocky Mountain Biological Laboratory long-term warming experiment.¹



Figure 2. Photographs of the Rocky Mountain Biological Laboratory experimental warming plots in summer (left) and winter (right). Photos courtesy of John Harte.

carbon into the soil because of their high photosynthesis rates. Earlier snowmelt created a longer dry period in late spring that the wildflowers couldn't cope with. So we identified the mechanism behind the feedback, quantified it, and went on to make predictions for other habitats, using a mathematical model of the carbon cycle.

BSJ: We read about your work to develop predictive models for ecological systems using the maximum information entropy (MaxEnt) method. How does this method work?

JH: There's a subfield of ecology called macroecology that's all about patterns in the distribution and abundance of species. Up through the 1990s, ecology had been gathering more and more beautiful data from censusing, and patterns were emerging. The question is, how do you explain and predict them? We'd like to have a "Grand Unified Theory" of macroecology that would explain patterns at all different scales, habitats, and species. That's a tall order. The common approach was focusing on identifying driving mechanisms of the patterns. Hundreds of mechanisms have been proposed as important in ecology: pathogens, herbivory, and so on. How do you make a model when you have so many mechanisms to choose from? So, I turned back to my physics roots. Thermodynamics and statistical mechanics are concerned with metrics such as the distribution of the speeds of molecules in a cylinder of gas. I realized that there was an approach to measure the information content in such a probability distribution called Shannon entropy. It was called entropy because it has a similar mathematical form to

the entropy function in physics. A physicist named Edwin Jaynes took the Shannon entropy function, maximized it subject to certain constraints, and was able to infer the shapes of probability distributions. Jaynes was a Bayesian statistician, so he used prior knowledge and acquisition of new data to upgrade predictions. When he maximized the Shannon information entropy of all the distributions in statistical physics, he was able to re-derive all of the results of statistical mechanics and thermodynamics. People then realized that they could use the same idea to derive distributions in economics, linguistics, neural net structure, and forensics. The method has been given a nickname—MaxEnt—for maximum entropy.

BSJ: How did you apply the MaxEnt method from thermodynamics to ecology?

JH: When Jaynes derived thermodynamics from MaxEnt, he used pressure, volume, and temperature as constraints. From those macroscale characterizations, he derived all the distributions at the microscale. In ecology, we have the total number of species on a hectare, the total number of individuals, and the total metabolic rate of the hectare of species—those are the analogous constraints. If we use those constraints and maximize information entropy of a certain well-defined probability distribution, we can calculate the shape of the distribution. Then we make predictions and compare them with the census data. It turns out that it works spectacularly. Interestingly, however, the theory breaks down when you apply it to ecosystems that are changing rapidly because of disturbance. This

could be either human or natural disturbance—anything that causes the system to start changing year by year. We hope the new theory we’re building, which includes disturbance, will actually explain the patterns better.

BSJ: How are you augmenting the static MaxEnt model?

JH: The original theory is a purely information theory approach—we do not assume any mechanisms. All we use as input are what we call the state variables: total number of species, total number of individuals, and total metabolic rate. Those are the result of mechanisms, of course, but we don’t ask what the mechanisms are—we just use the variables as constraints for the patterns. In a disturbed system, we have to understand how the state variables are changing, so that requires explicit introduction of mechanisms. One such critical mechanism is migration from outside, so we can disturb the system by altering the parameter that describes the immigration rate. Other mechanisms include the birth, death, and growth of individuals, and the extinction of species. Now it becomes a hybrid theory, because it has both the MaxEnt component and the mechanistic component. We are quite sure that we can’t develop a purely statistical MaxEnt theory of disturbance, because disturbances are all unique and responses critically depend on the mechanisms that create the disturbance.

BSJ: What are some of the most promising applications of the MaxEnt Theory of Ecology?

JH: So far, the most promising applications have been to accurately predict all of these patterns that are in the literature—to explain puzzles that have perplexed ecologists for decades: why do distributions and relationships look the way they do? If we can develop a successful MaxEnt-mechanism hybrid theory, it will be very useful for predicting things like the fate of ecosystems under human disturbance. Physics has great prestige today because we have theory that explains pattern. Ecology is still largely in the stage of lots of observation, with little understanding of the origins of patterns or ways to predict them. The Maximum Entropy Theory of Ecology is an attempt to give ecology more credibility. If astronomers have calculated that an asteroid is about to hit Earth and they go before Congress, no one would doubt the astronomers’ calculations because they’re based on good, solid theory. The physicists have credibility. If ecologists go to Congress, as we’ve done, and say a different kind of asteroid is hitting Earth: the global extinction of a sizable fraction of all the species on the planet, the

“Helping to build a foundation of solid ecological theory will give ecologists more credibility in the policy arena.”

response is typically, “Why should we believe you?” Helping to build a foundation of solid ecological theory will give ecologists more credibility in the policy arena.

BSJ: A lot of your work intersects with human population studies. How do we ensure food security into the future?

JH: A high priority of all governments should be to confront the problem of population. Our numbers are growing, and that is going to compound every other problem that we have, from climate change, to drought, to soil erosion, and so forth. This is partly a human rights issue. There are a few hundred million women around the world who don’t have access to contraception. A combination of religious and political ideology has denied women a fundamental right, which is to control their own reproduction. Providing women everywhere with the means to implement family planning, if they so choose, would be the single most important step to ensure food security. There are many other things, too. Just as with income inequality, we have food access inequality. But all of these other problems are more and more difficult to solve when there are more and more people. Achieving sensible, workable governance under conditions of overpopulation is very difficult. So I would put the population focus at the top, but I would also add dealing with inequality and with land use management. Some of the farming techniques that would avoid soil erosion, improve yields, save water, produce more healthy food, and promote biodiversity, would also sequester more carbon and help deal with the climate issue. There’s a synergy in the solutions to most of the problems that humanity faces. And no one solution, alone, will be enough.

BSJ: What are some future directions of your research?

JH: Right now, all of the focus is on extending the static MaxEnt theory—the one that worked in undisturbed systems—to systems that are changing rapidly because of disturbance. That’s a massive project, and by no means have we completed it. That’s the focus of ongoing work—turning a static theory into a dynamic theory.

BSJ: Thank you very much for your time!

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ADENO-ASSOCIATED VIRUSES: VEHICLES FOR RETINAL GENE THERAPY

An Interview with Professor John G. Flannery

By Arjun Chandran, Cassidy Hardin, Michelle Lee,
Melanie Russo, & Elena Slobodyanyuk

Dr. John G. Flannery is a Professor of Vision Science in the School of Optometry and a member of the Neurobiology Division of the Department of Molecular and Cell Biology at the University of California, Berkeley. Professor Flannery's research centers on developing viral-mediated gene therapies for inherited retinal diseases. In this interview, we discuss the optimization of adeno-associated viruses (AAVs) for retinal gene therapy and their applications to optogenetic vision restoration.



Professor John G. Flannery

BSJ: How did you get involved in the field of neurobiology and studying inherited retinal degeneration?

JF: I was a resident assistant in the dorms at UC Santa Barbara when I was an undergraduate. There was a rule that you couldn't stay for the summer semester unless you took classes, and I didn't find any of the summer classes interesting. So, I volunteered in the lab of a new assistant professor who had just come from Johns Hopkins University, and he worked on the retina. It was total random luck!

BSJ: What are some of the main causes of retinal degeneration?

JF: They're all genetic. The first gene was found in the 1980s: a point mutation of rhodopsin was responsible for a form of retinitis pigmentosa, an inherited eye disease. That was when I was a graduate student. Now, there has to be a webpage of all the disease-related genes because there are about 250 different ones. Retinal degeneration is one of the most heterogeneous diseases. On the upside, it looks like the curve is plateauing; we aren't finding many more new genes. But there are always going to be new ones because people can always get new mutations. Fortunately, it looks like the number of causative genes is going to remain at just a couple hundred.

BSJ: Why were AAVs chosen as an optimal vector for retinal gene therapy?

JF: Most known viruses only infect dividing cells. Because your nervous system cells don't divide after you're a fetus, you have to have a virus that will infect non-dividing cells for any neurobiological use. So that narrows down the number of known viruses. You also want something that lasts for a long time. Most of the patients come into the clinic when they're teenagers, so you need something that persists for their entire lifetime. Some of the viruses don't put the gene into the host cell nucleus, so they only last for a couple of weeks. This defeats purpose of doing gene therapy because in that case, you can just inject the missing protein directly. Now the number of candidate viruses goes down to a handful. The adeno-associated virus has a very simple genome and is easy to engineer. 80% of people in the world are carrying the AAV2 serotype already because it doesn't cause disease and is easy to get from something like an elevator button. The downside of AAV is that it doesn't package a very big gene, so you can only put five kilobases into it.

BSJ: What are some strategies you have used to try to overcome AAV packaging limitations?



“We would like to design treatments that are one-time injections that are affordable for all patients.”

JF: People have tried to cut the causative gene into pieces and put it into two viruses. If you look at the sequence of the protein and find some intron-exon boundaries, you can cut it and put it into two vectors and have the cell make the complete protein. However, this protein expression is very low, around 10% of what you would achieve if you put the entire protein coding sequence in a single AAV. The concentration of the AAV viruses that we make in the lab is in the hundreds of millions per milliliter. If you need to put in two different viruses to make a single product, a cell has to get infected by both viruses. It also has to splice the gene together. It's just not a very efficient process. People have made significant improvements in the junction between the two halves of the gene, but the cell functions better with a full-length gene in one place in its chromosome, not divided into two locations.

BSJ: How are rational design and directed evolution used to improve the AAV vector?

JF: AAV occurs in nature; it wasn't made by scientists. It's been around for millions of years and has been optimized by evolution for many properties, some that we don't necessarily want for use in the eye. It has evolved to be really good at infecting the liver, kidneys, or some other tissue, but we want it for properties that are unique to the retina. The chances that the wild type AAVs are optimal are low, so we use these new methods to try to improve AAV properties. Mainly, we want two things. Almost all the gene mutations affect photoreceptors, not other retinal cells, so we need the virus to penetrate through the first layers of the retina, the ganglion and bipolar cells. We don't want to put the virus directly near the photoreceptors, because we would have to put the needle through the patient's retina. Second, the virus has to infect the photoreceptors after it gets to them. Those are the two things that our screens look for. Rational design assumes that you know something about what you want. We haven't done much rational design because we don't really know what receptors to look for or how to best reach the photoreceptors in the retina. With directed evolution, we make every possible change we can on the outside of the virus and then screen the variants. Most of the ones we make are not very good, and a lot of them are worse than the ones that occur in nature. But since we make so many—around 100 million variants—and we have a way of screening them, we've been able to make some vectors that are much better than wild type AAV.

BSJ: What is the mechanism by which AAVs deliver genetic cargo?

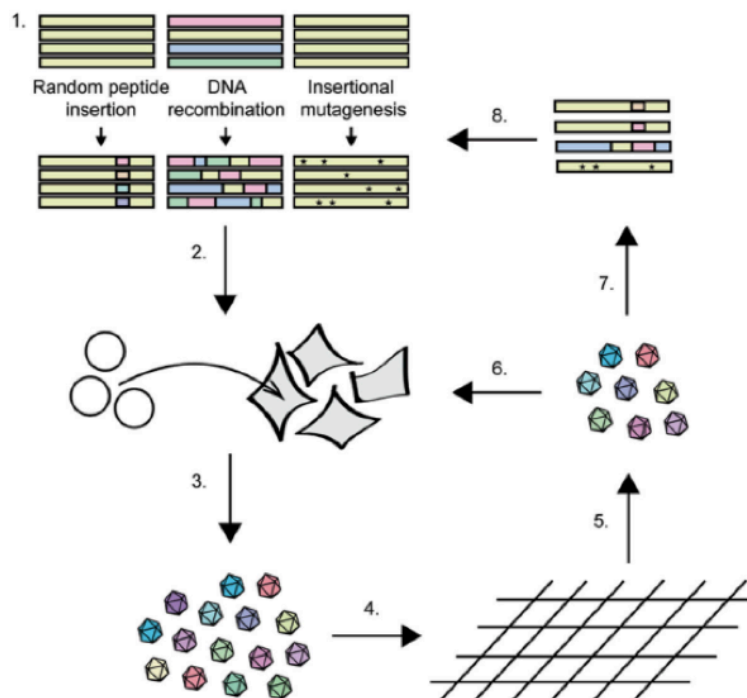
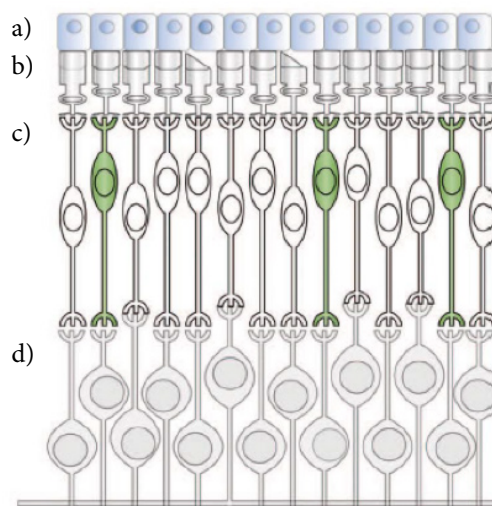


Figure 1. Steps in the directed evolution of AAV. Viral variants are created with different mutations (1) and packaged into cells (2). AAV is harvested (3) and subjected to selective pressures (4). Successful viruses are isolated (5) and undergo further rounds of selection (6).¹

Figure 2. Schematic of the retina featuring the retinal pigment epithelium (a), photoreceptor cells (b), bipolar cells (c), and ganglion cells (d). The optogenetic treatment targeted bipolar cells, which are highlighted in green.²



JF: The cell has a surface receptor, and AAV sticks to that and gets into the cell. If your cell doesn't have that receptor, the AAV just bounces off. In the nervous system, the Müller glia in the retina don't have those receptors, so the virus doesn't infect any glia. After it gets recognized by the surface, the virus goes into the cell. The cell has great mechanisms for taking off the outside capsid, and then the virus gets into the nucleus. That's the property of a molecular syringe that we really need, because if you don't use a virus and put the therapeutic DNA outside the nucleus, the cell will destroy the invading DNA. Viruses have evolved really good tricks for getting into the cell, protecting the target cargo, and entering the nucleus. We need to keep all of those things so that the therapeutic gene gets into the cell's nucleus and stays there expressing the therapeutic protein for a long time.

BSJ: We read about your recent study using AAVs to restore retinal function in retinitis pigmentosa mouse models. Why did you choose to target the Crumbs protein complex?

JF: There are two kinds of Crumbs diseases. One develops relatively fast—the children present symptoms in their first couple of months. Their vision is quite poor and deteriorates quickly. The other type presents itself in teenagers and people in their early twenties. What does the Crumbs protein complex do? The Crumbs complex is the junction between glia and photoreceptors, so it makes a barrier between the inner and outer retina and is critical for the retina's lamination. In children with early-onset severe Crumbs, their retina is twice as thick as normal because the retinal cells can't make well-ordered layers. I work for the Foundation Fighting Blindness, and we are approached by donors all the time. A group of families asked us to try and put the Crumbs proteins back, and they raised money to fund a clinical trial. The Crumbs disease was picked because we have the tools to work on its degeneration right now, even though it is not the most common defect.

BSJ: In the Crumbs study, you found that CRB1 protein expression triggered an immune response in some cells. How can you explain this response given the low immunogenicity of AAVs?

JF: The immune response is due to the Crumbs protein, not the AAV. It's a recessive disease,

so the patients have never made the protein before. Because the Crumbs protein is located on the surface of the cell, the immune system is able to mark it as foreign when the patient starts making it. For patients who have null mutations for proteins that stick out of the cell or are secreted, a big question is whether there will always be an immune response, and whether it can be managed. But we know that it's not the AAV that's immunogenic; it's the novel protein you're making.

BSJ: We also read about your work to develop optogenetic gene therapies using photoreceptive G-protein-coupled receptors. Can you explain the concept of a retinal prosthetic? What cells did you choose to target for this treatment?

JF: In the case of Crumbs or such diseases where there's one defective gene that we're able to replace, that's certainly the best thing to do. However, once the cell is gone, putting the gene back is not going to help. For patients who have lost all their photoreceptors, they still have bipolar and ganglion cells, and light normally goes through those. Through imaging, we can tell that patients have those cells

“The idea of a retinal prosthetic is to put in a gene that adds a new function to the cell rather than replace a missing gene.”

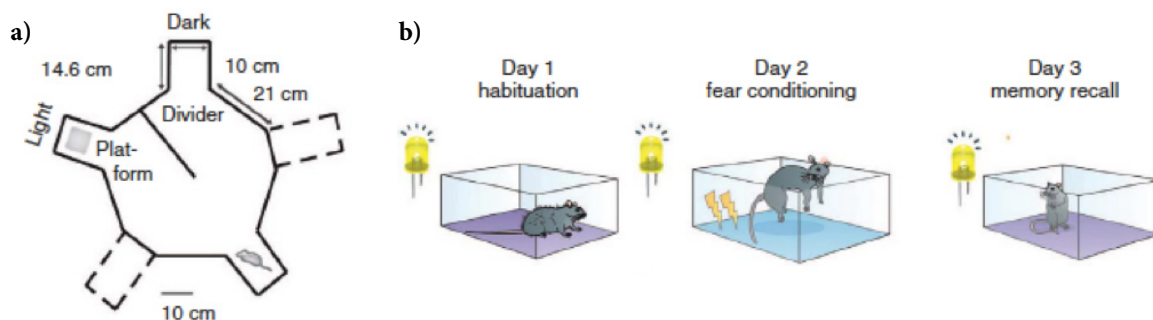


Figure 3. a) Diagram of the water maze used to evaluate visually-guided behavior of optogenetically treated mice. b) Visually-cued fear conditioning of mice revealed that optogenetic treatment restored their ability to distinguish light patterns.²

for decades after they're blind, but those cells only respond to glutamate, not light. The idea of a retinal prosthetic is to put in a gene that adds a new function to the cell rather than replacing a missing gene. For example, you can make the cells light-sensitive. You put a photoswitch gene into the virus, target the virus to bipolar or ganglion cells, and express the optogenetic proteins. They won't be as good as a normal retina because there aren't as many bipolar and ganglion cells as there are photoreceptors, but you can still treat patients who have lost their photoreceptors.

BSJ: What is the main advantage of using rhodopsin as the light-sensitive protein for this therapy as opposed to bacterial opsins such as channelrhodopsin?

JF: What's great about channelrhodopsin is that it's one protein—an ion channel that opens and closes with light. It's super fast but not very sensitive because the sensor and the channel are the same protein. With channelrhodopsin as the actuator, the light has to be the intensity of Miami Beach at noon to elicit a response in the retinal cells. The other problem is that channelrhodopsin doesn't have any light adaptation. If the intensity is below that really bright light, channelrhodopsin activity doesn't decrease, it stops. In normal vision, light sensitivity covers many log units of brightness, from twilight to the bright outdoors. For channelrhodopsin, the sensitivity is narrow: one log unit. With rhodopsin, it's a light sensor, but there's also a cascade of other enzymes linking it to a separate ion channel. With this you get tens of thousands-fold amplification and also adaptation. When we used rhodopsin and thus separated the sensor from the ion channel, we got four log units of adaptation and about ten thousand times more sensitivity than channelrhodopsin.

BSJ: Can you tell us about the behavioral experiments you conducted to evaluate retinal functioning in mice?

JF: We used a water maze with a target. We have LED lights on two platforms and a little table that's slightly under the surface of the water. If the mouse is able to tell the difference between the flickering and steady light, it can get out of the water onto the platform. The untreated blind mouse will swim around but is not able to see the target. However, the treated mouse and the wild type mouse will quickly learn to swim straight to the target. Another experiment used a box with metal bars on the floor that give a very mild shock. We wanted to see if the mice can tell the difference between vertical and horizontal bars to avoid the shock. The untreated mouse with no photoreceptors spent half the time on either side, whereas the normal mouse very quickly learned the task. The optogenetically treated mouse could also tell the difference and complete the task. Then, we asked if the treated mouse could tell the difference between moving bars and bars of different width. It could do all of that. For the newest task, we built a box with toys. If you put a ball, a triangle, and anything else with the normally-sighted mouse and track where it goes, the mouse explores all the toys. The blind mouse just goes to the first one it finds and stays there because it doesn't know there are any other toys there. It's perfectly happy with the first toy, just sensing it with its whiskers. The optogenetically treated mouse pretty much explores as much as the wild type mouse.

BSJ: Going forward, how do you see AAV gene therapy evolving?

JF: We'd like the price to go down. There's one therapy that just got approved for Leber's congenital amaurosis, but it's priced at \$500,000 per eye. There are about 500 patients in America who have that, and there are millions of patients who have mutations of the other 250 genes. We want to develop therapies for more patients, for bigger disease genes, and for dominant diseases. We would like to design treatments that are one-time injections that are affordable for all patients.

BSJ: Finally, what are some future directions of your research?

JF: We'd like to get optogenetics to the clinic. It's great that the first AAV therapy got approved in January. If you talk to biotech and pharmaceutical companies, they have this phrase called the "Valley of Death." It typically costs 500 million dollars to run any drug trial through the FDA, from the lab to the clinic. Sometimes it's twice that. It's not really the issue of the FDA to worry about the price—they're worried about not hurting anybody. In this "Valley of Death," there's so many things that can look like they're working in the lab that never get to the clinic because they fail in testing, or there's some side effect, or you just don't have the money. Part of the reason the first treatment for Leber's congenital amaurosis from Spark Therapeutics costs \$500,000 per eye is that there's not very many patients, so the number of dollars a company needs to charge per patient to break even is going to be big. We hope that the technology advances, there will be more therapies for diseases that have more patients, and the price will be not crazy.

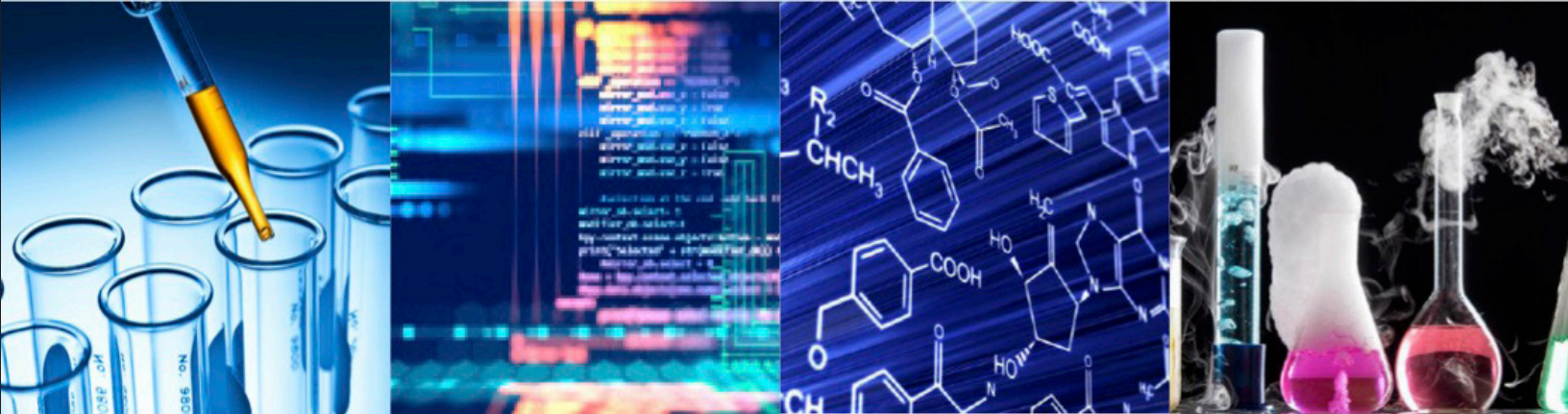
BSJ: Thank you very much for your time!

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TIPS ON SCIENCE CAREERS:

YOUR GRADUATE STUDENT INSTRUCTORS SHARE WISDOM

BY ARJUN CHANDRAN, NIKHIL CHARI, CASSIDY HARDIN, MICHELLE LEE, ROSA LEE, PHILLIP DE LORIMIER, MOE MIJJUM, MELANIE RUSSO, SONA TRIKA

We asked nine UC Berkeley graduate student instructors (GSIs) to share their post-graduate experiences and give advice to undergraduates who want to pursue a career in science.

BSJ: What has sparked your interest in pursuing graduate school in your field?

AARON BROOKNER: I was always split between math and programming. I knew that I would either go into a math graduate school or software engineering. There was a summer where I did both: I participated in a math research program and a programming internship. While I was doing math, the startup that I was interning for went bankrupt. I had previously decided that summer would guide me to the right choice and it unambiguously pointed to math.

BSJ: How has your perspective on science changed after you entered the world of research?

AARON BROOKNER: Honestly, my perspective on science has not changed at all.

AMANDA KELLER: My first reaction is that there is so much reading. All I do is read. I had no idea about this. You will read 50 articles, and then your project will change completely. Also, I wouldn't say that you get that much freedom in what you want to study—it depends on what funding you're able to get, and for the most part that is out of your control. After I got over my initial shock of how much reading there was in research, I noticed that a lot is lost in re-

search translation. What we do in the lab often doesn't see anything other than the scientific journal that we publish in.

MAZEN HAIDAR: I learned that coming up with results or answers is not always evident or possible. There are times when “no result” is the answer to the research question.

ELIE ALCHEAR: When I first started research, I wanted to get a Ph.D. and become a professor. After a while, I learned that there was a lot of BS involved in the world of academia. Some reviewers seem to care more about how you referenced a previously published paper than about whether you are right or wrong. I also realized that some papers are more a compilation of previous works than anything original. At least in the field of engineering, the impact of professors is evaluated based on how many papers they publish. Consequently, they care more about quantity than quality. This became one of the reasons why I didn't want to continue academia. Because of these issues, I want to finish my masters and become more field-oriented.

ROBERT-MARTIN SHORT: I understand more now about how papers are published—the politics that goes behind publishing papers. But I think science is now more of a human endeavor to me than before. There are problems to be solved, the peer-review process. Before, as an undergrad, I just thought: “Oh, this is a published paper, therefore it must be right.” But now, as someone who's seen the process more, I'm a bit more skeptical. If something in the literature you read sounds like garbage, it probably is garbage—but some things are good. In academia, you learn what to believe, and what to take with a grain of salt.

“I think that the best science happens when you have independent ideas but work collaboratively.”

MELISSA HARDY: When I first came here, it was somewhat eye-opening to see how fast-paced research is at Berkeley compared to at my liberal arts undergraduate college.

LINDSEY HENDRICKS-FRANCO: I think the thing that I've learned and continued to re-learn over the course of my research career is that at its core, it's a problem-solving process. You have to have the flexibility to confront the individual challenges that come up with any given project and solve them while maintaining the scientific integrity of your work—I think that's what unites all research.

NEHA MITTAL: I appreciated science more from a product perspective before, but now I respect research more as a way to innovate. I don't think innovation can occur just via startups. A large part of innovation is based on people who have the patience and courage to go into something completely new and become an expert in it.

BSJ: What are the most important skills that students need to be successful in your field?

AARON BROOKNER: UC Berkeley has a graduate preliminary exam that actually just tests undergraduate knowledge. So, in the field of mathematics, I think a solid understanding of undergraduate math really helps.



Figure 1. Neha Mittal, Master of Information Management and Systems (MIMS) student at the Berkeley School of Information.

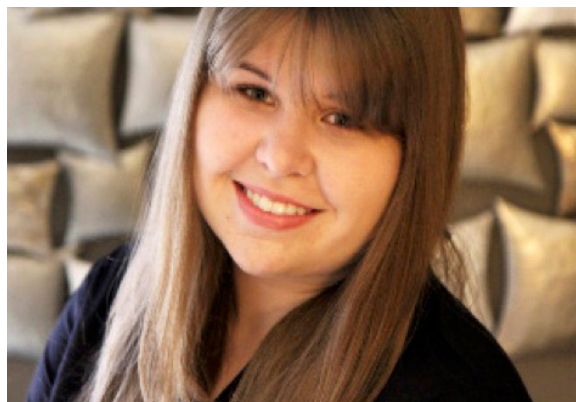


Figure 2. Melissa Hardy, graduate student in Chemistry at the Sarpong Lab in the UC Berkeley College of Chemistry.

ELIE ALCHEAR: Ideally having some field-applicable research, especially in civil engineering, would be great. It not only helps resume-wise, but it also aids in personal growth and helps you make up your mind when it comes to what aspect of civil engineering you want to specialize.

AMANDA KELLER: Toxicology is bench heavy, but I always say that lab skills are something that can be taught. What I think you really need is to be resilient. Science is 99% failure and if you fail and cannot get back up on your feet, this discipline will destroy you. You need to know that if your experiment failed, it wasn't your fault—it was science. If it was your fault, you need to get up and try again. You can teach lab skills but you cannot teach someone to be resilient. Also, creativity is huge in science. I was once asked to come up with 20 new and different project ideas. You have to couple creativity with critical thinking skills to come up with something novel.

PHILJUN KANG: I did my master's degree in organic synthesis, so I have a very strong synthetic background, but that's it. Especially in materials science, you have to know a lot of different things. Sometimes I don't know the value of my materials, because I just focus on their synthesis. But others know that these materials can be very precious, if used in a certain field. So I think, these days, discussion and openness to other researchers are very important to improve prospective knowledge in science.

MAZEN HAIDAR: Students need to be patient; research can be long and even frustrating in certain cases. The other skill I would emphasize is integrity. Referencing others' works or ideas is a requisite to any successful research paper. In addition, just like coursework, research requires time-management skills, organization, and self-discipline. Last but not least, students must be passionate about the topic they are working on: there is no success without passion.

ELIE ALCHEAR: I would put most emphasis on soft skills—communication skills, ability to present, public speaking, and so on—and creativity.

ROBERT-MARTIN SHORT: My field is geophysics and maybe a bit of computational geophysics. In terms of hard skills, you definitely need computer skills—know the Unix command line and maybe also a programming language. In terms of soft skills, you need to be independent in your research, be able to cooperate with people on research projects, and finally, present your work well. Usually, when students go to conferences and workshops, they are asked to stand up and talk about what they have done. So, that's a skill on its own—you can have good research but if you don't talk about it well, then nobody is going to understand and appreciate it.

MELISSA HARDY: In my opinion, persistence is by far the most important trait for successful graduate chemists. Inevitably, there will be times when research doesn't work as you expect it to and your project can get derailed in unexpected ways. Per-



Figure 3. Philjun Kang, graduate student in Chemistry at the Yagi Lab in the UC Berkeley College of Chemistry.

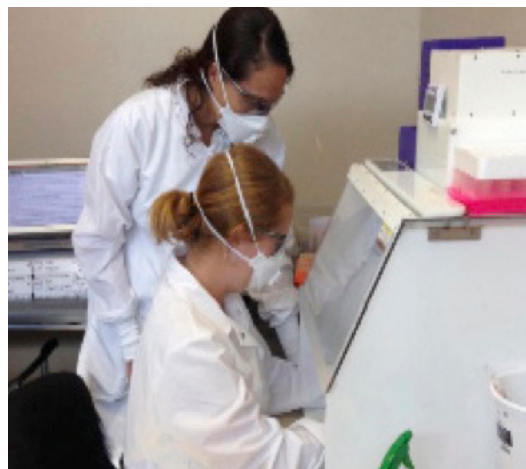


Figure 4. Amanda Keller (front), graduate student in Toxicology at the Smith Lab and Dr. Fenna Sillé (back), Johns Hopkins University professor.

sistence, often coupled with passion, will keep you motivated and moving forward even when the chemistry has a mind of its own.

NEHA MITTAL: One core skill, though this might be for an engineering mindset specifically, is it's important for people to get their hands dirty with whatever they're doing. It's important to get involved in projects and even project management, if possible.

BSJ: Do you have any advice for undergraduates?

AMANDA KELLER: As an undergraduate, I worried so much about my future. I was like: "I need to do this, I need to study this..." I wish I had studied abroad instead, which is why I went to Guatemala after graduating. I wish I had known what I know now, to just breathe and live life.

LINDSEY HENDRICKS FRANCO: I encourage anybody who wants to get into research to start by doing some supervised research, where you just work on a graduate student's or a professor's project, and then work toward planning an independent project for either your junior or senior year.

BSJ: What are the biggest lessons you've learned as a graduate student at Berkeley?

LINDSEY HENDRICKS-FRANCO: I think that the best science happens when you have independent ideas but work collaboratively. You can come up with some very creative ideas by having a conversation with other people who are talented and motivated and know a lot about the subject that interests you. Also, make sure to interact with the public. I've enjoyed the opportunity to do interviews with science journalists locally. It's the right thing to do to share what you're learning with the public. The more that you practice it, the better that you get at understanding the public's concerns.

IMAGE REFERENCES

All photos of individuals courtesy of the respective graduate student pictured.



The Mysterious Loss of the Third Molar in the New World Monkey Family *Callitrichidae* and its Relationship to Phyletic Dwarfism

BY JEFFREY L. COLEMAN

ABSTRACT

Callitrichidae is the smallest of five families of New World monkeys, descendants of African simians that colonized South America around 40 million years ago. Callitrichids, unlike other primates, do not develop third molars. It has been proposed that this is due to a crowding out event, whereby, over evolutionary time, a shortening cranium associated with decreasing body size pushed out the last molars on a tooth row that did not shrink as rapidly. This would create an equivalent relative area for chewing compared to non-callitrichid New World monkeys. After analyzing craniodental measurements of 143 New World monkey specimens, I found that callitrichids display an especially short postcanine row to body size ratio. A direct link between reduced body mass and third molar loss in callitrichids remains unsubstantiated, although current research suggests they may be related via an underlying genetic mechanism. Researching these associated genes may have implications for human health.

INTRODUCTION

Among the five New World monkey, or platyrrhine, families, which include Aotidae, Atelidae, Cebidae, Pitheciidae, and Callitrichidae, only the last is believed to have experienced phyletic dwarfing, or decreasing in size over evolutionary time.¹ Research on phyletic dwarfing has usually concerned islands. However, its occurrence on larger land masses may be a more common phenomenon than much of the literature suggests, with evolutionary episodes happening independently in a few mammalian lineages besides callitrichids.² Reduced body size is part of a suite of distinctive features that define callitrichids: higher rates of twinning, the presence of claws, and the absence of third molars and hypodemes also distinguish this family.¹ However, due to an incomplete fossil record and relatively recent resolution in the phylogeny of platyrrhines, it is unclear whether these characters evolved synchronously during one evolutionary event or during parallel episodes of body size reduction. Either way, morphological diversification in callitrichids appears to have resulted from a strong positive selection on size, so size is an adaptive trait in this family.²

The sparse fossil record of callitrichids from the Plio-Pleistocene has limited the scope of research on the climatic and ecolog-

ical conditions that were prevalent during the evolution of their dwarfism. Early research in this topic posited that no minimum date can be set for the occurrence of one or several dwarfing episodes. However, a likely location for these events, the forests of continental South America, is supported by the current and likely historic habitat distribution of callitrichids. While larger platyrrhines are found principally in gallery forest, the smaller ones, particularly the callitrichids, are found in dense secondary growth and scrub forest. Possibly, times of dryness and of reduced and widely separated forest regions, in concert with the normal perturbations of the tropical environment, may have led to limitations of resources. This may have favored smaller species that would rely on less food and territory, then allowing the callitrichid population to boom after these resources replenished. Tooth morphology could have diverged as an adaptation to dietary specialization either during or subsequently to this period.²

It is possible, then, that understanding the postcanine dentition of callitrichids can shed light on an evolutionary relationship between their dwarfism and third molar loss. The focus specifically on postcanine dentition, the molars and premolars, is appropriate, as its genetic regulation and development are separate from those of anterior dentition.³ As well, postcanine dentition is known

to be correlated with body size across primates.⁴ Generally, dwarfed lineages have been shown to exhibit different ontogenetic scaling, the change in the size of one feature relative to another during early evolutionary and embryonic development, from the normal interspecific trend.⁵ Gould⁶ suggested that in dwarfed lineages, body size decreases far more rapidly than the postcanine dentition. For example, an increase in relative molar size in Pleistocene marsupial lineages was associated with decreasing body size.⁷ This scaling phenomenon, whereby an organ grows or shrinks more slowly than the rest of the body, in a lifespan or over evolutionary time, is known as negative allometry.¹ Callitrichids, not unusual among primate families in their tooth eruption patterns at birth, were assumed to follow this trend.⁸ Hence, Gould hypothesized that the oversized molar battery relative to a reducing mandible may have led to the crowding out and loss of their third molar. This would allow for a functional chewing surface that takes up a portion of their skull that is roughly equivalent to non-callitrichid platyrrhines. A 1993 study suggested, on the other hand, that the postcanine row of callitrichids takes up significantly less area of the entire cranium than that of non-callitrichid platyrrhines in three of four genera tested, the exception being in the *Leontopithecus rosalia* species.⁹ A small handful of species that dwarf have been shown to lack this negative allometry.⁷ If callitrichids do not fit this trend as well, it may be related to their rare change in ontogeny, or development, whereby prenatal growth rate was slowed down. This would explain how their body size was reduced.¹⁰ While developmental genetics has been shown to inform craniofacial and dental morphology across a variety of mammalian species,¹¹ the type of body size reduction seen in callitrichids is likely a rarity among mammals apart from domestic dog breeds, among which postnatal growth rates vary little.¹² Further, it has been suggested that dwarfism in callitrichids is associated singularly with these changes in prenatal growth rates rather than the duration of gestation, postnatal growth duration, or postnatal growth rates. The exception to this is *C. pygmaea*, whose extremely small body mass is probably caused by a lagging of both prenatal and postnatal growth rates, suggesting that their accelerated sexual maturation relative to the rest of their development could have played a role in the evolution of this species.²

I hypothesized that the loss of the third molar in callitrichids is correlated with a shortening of the facial skeleton that over evolutionary time, forced out the last molar. This would be consistent with the crowding out hypothesis and the general trend in allometry seen in dwarfing lineages. I also hypothesized that callitrichids' unique ontogeny would be irrelevant in affecting this ubiquitous trend. Thus, there would be no significant difference between the relative size of the postcanine tooth row, both maxillary and mandibular, of callitrichids when compared to non-callitrichid platyrrhines. The Plavcan and Gomez study that rejected this crowding out hypothesis tested a sample of 18 species, including four callitrichid species. This represented a smaller species richness than the 42 species, including five callitrichid species, used by my study. I suspected that the lower species di-

versity of the Plavcan and Gomez study may have limited its comparative power, with the postcanine row trend only becoming visible when platyrrhine families are studied more broadly.

MATERIALS AND METHODS

Materials

To evaluate relative postcanine tooth row sizes between callitrichids and non-callitrichid platyrrhines, I compared ratios of postcanine row length to various measurements of cranial length between these two groups. Instead of measuring body volume directly, I compared the tooth row with skull measurements that are known to be strongly correlated with body size and which are more feasible to measure.¹³

I collected data from 157 platyrrhine skulls in total, comprised of 34 skull photographs of specimens from the Smithsonian National Museum of Natural History, 106 skulls from the Museum of Vertebrate Zoology mammalogy research collection, and 17 skulls from the California Academy of Sciences mammalogy research collection. Of these 157 specimens, the 143 adults, defined as having fully erupted dentitions, were analyzed. This way, I knew intraspecific variation would not be confounded by variable stages in development. The sample spanned all five families of platyrrhines, consisting of 57 callitrichids, 22 cebids, 14 aotids, 21 pitheciids, and 29 atelids. In total, there were 16 genera and 42 species included in the study.

Data Collection Methods

I measured the upper and lower postcanine tooth row lengths on both sides of the face, mandible lengths, cranium lengths, and calvarium, or skullcap, lengths, of every museum specimen. I used Mitutoyo digital calipers, and performed three separate trials for all seven measurements on each specimen (Figures 1-3). I conducted photograph measurements on ImageJ version 1.48¹⁴ with a standardized protocol, using the same landmarks as on the museum specimens. However, I did not gather calvarial measurements from the photograph specimens since dorsal and lateral views of the skulls were unavailable.

Analytical Methods

I generated all statistical analyses in R, version v3.1.2.¹⁶ I averaged the measurements from the three different trials for each specimen after determining that the differences in the measurements were not statistically significant. To resolve concerns about possible bias that could have arisen from the two different measurement methods, I ensured that photograph and caliper measurements gleaned from 10 randomly selected platyrrhine skull specimens from the Museum of Vertebrate Zoology were insignificant in their differences. I subsequently collapsed the two data sets (Table 1).

After verifying that postcanine row length side differences were insignificant, I averaged the two right and left postcanine tooth rows of the maxilla as an upper tooth row, and the two mandibular tooth rows as a lower tooth row. I calculated



Figure 1. Green line shows visual of cranial length measurement. I measured cranium on the ventral side of the skull, from the front of the alveolus between the central incisors to the midpoint of the occipital crest where the parietal midline suture begins. The blue line shows visual of postcanine tooth row length. I measured this from the most anterior point of the second premolar to the most posterior point of the most posterior molar, either second or third. Photograph represents a sampled specimen of the species *Leontopithecus rosalia*. Specimen number is 588174 from the Smithsonian Institution National Museum of Natural History.



Figure 2. Blue line shows visual of mandible length measurement. I took mandibular length as the path from the superior infradentale between the central incisors to the midpoint of the superior left condyle of the mandible (15). Photograph represents a sampled specimen of the species *Leontopithecus rosalia*. Specimen number is 588174 from the Smithsonian Institution National Museum of Natural History.

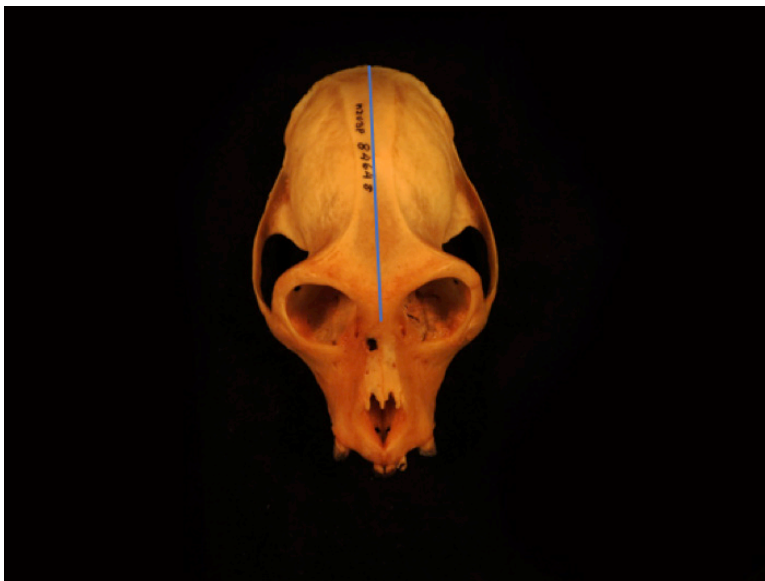
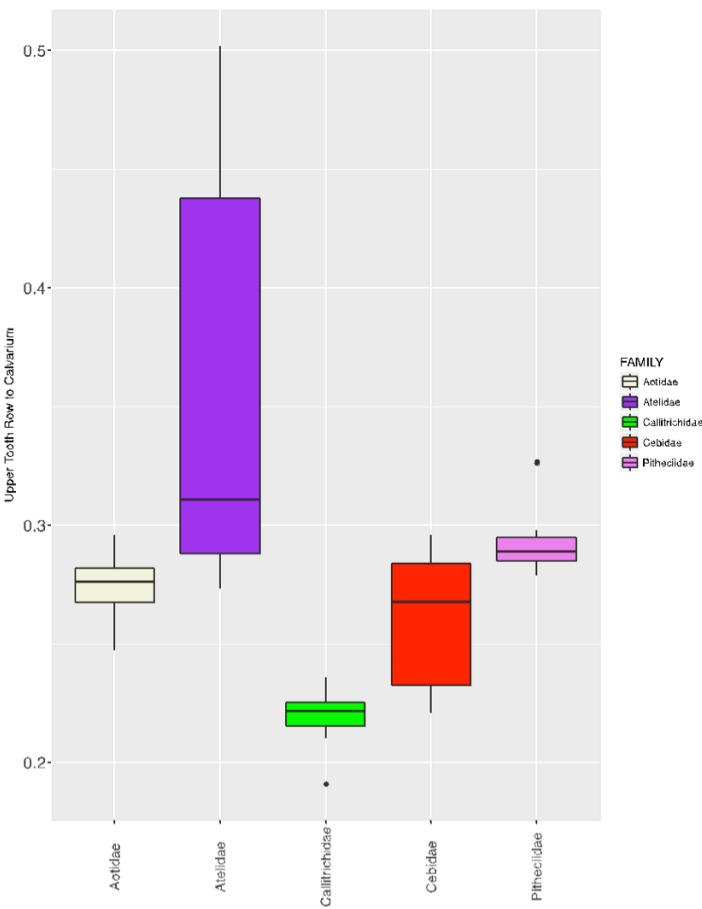


Figure 3. Blue line shows visual of calvarium length measurement. I measured calvarium length from the midpoint of the suture dividing the nasal and frontal bones to the midpoint of the occipital crest, the suture dividing the occipital and parietal bones, where the parietal midline suture begins. This was conducted laterally with the calipers placed parallel to the occlusal plane of the teeth rather than perpendicular as in the other measurements. This was due to its ability to generate reproducibility and consistency given the shapes of the skulls. Photograph represents a specimen of the species *Alouatta guariba clamitans* not sampled in this study. Specimen number is 84648 and is from the Zoology Museum of the University of São Paulo.

Family	Statistic	Right Maxillary Postcanine Length (mm)	Left Maxillary Postcanine Length (mm)	Right Mandibular Postcanine Length (mm)	Left Mandibular Postcanine Length (mm)	Mandible Length (mm)	Calvarium Length (mm)	Cranial Length (mm)
<i>Aotidae</i>	Sample Size	11	12	11	12	12	14	14
	Median	14.13	14.10	15.93	15.82	38.34	51.19	60.19
	Mean	14.20	14.16	15.82	15.77	38.71	51.73	60.29
	SD	0.52	0.45	0.44	0.46	1.24	2.20	1.54
<i>Atelidae</i>	Sample Size	20	20	25	25	29	29	29
	Median	24.30	24.81	28.11	28.01	72.64	79.81	106.13
	Mean	26.63	27.60	30.43	30.91	75.68	78.96	108.54
	SD	4.29	5.01	5.69	5.86	9.61	4.94	6.56
<i>Callitrichidae</i>	Sample Size	53	52	52	54	53	23	56
	Median	8.98	8.91	10.38	10.44	30.70	41.19	44.29
	Mean	9.36	9.35	10.74	10.73	31.01	39.36	45.30
	SD	1.65	1.63	1.79	1.74	4.17	5.26	5.88
<i>Cebidae</i>	Sample Size	12	15	17	16	22	21	22
	Median	16.31	20.38	23.75	22.92	55.08	68.91	84.79
	Mean	16.59	17.71	20.48	20.15	49.66	66.34	80.36
	SD	4.29	4.35	4.80	4.80	12.04	10.25	14.12
<i>Pitheciidae</i>	Sample Size	17	15	18	16	19	21	21
	Median	15.30	15.44	17.44	17.37	41.53	53.22	62.14
	Mean	16.86	16.94	18.91	18.96	47.40	56.44	68.99
	SD	2.31	2.47	2.81	2.96	9.81	7.21	12.37

Table 1 (top). Sample sizes and descriptive statistics for all traits measured for all five families.

Figure 4 (bottom). Boxplot shows ratios of upper postcanine tooth row lengths relative to calvarium lengths across families. Upper and lower postcanine rows compared to all cranial measurements showed similar trends, with callitrichids having relatively smaller postcanine tooth rows than all other families.



the ratio of the upper and lower tooth rows to the mandible length, calvarium length, and cranium lengths to standardize the tooth row data. I ran correlations to see how strongly the average upper and lower tooth row lengths were correlated with mandible, calvarium, and cranial lengths, and to determine how strong the correlations among mandible, calvarium, and cranial lengths were. Since all skull measurements are related to body size and were expected to be highly correlated with each other, the latter correlations acted as a control.¹³

With the callitrichid species *L. rosalia* and *C. pygmaea*, I performed separate ANOVA analyses. In one, I treated *C. pygmaea* as a family to compare its ratios to those of all the other callitrichids, as well as to those of the other platyrrhine families. This was to examine whether their unique ontogeny and even more dramatic dwarfing may be associated with significant negative allometry even relative to other callitrichids. In the other, I did the same for *L. rosalia*, also to learn if they displayed uniquely strong negative allometry. This was based on the findings of Plavcan and Gomez that *L. rosalia* was the only callitrichid to not display a relatively smaller postcanine row compared to non-callitrichid platyrrhine species.^{2,8} In this case, I wanted to investigate the possibility that only this genus's lineage shows negative allometry. Besides these examples, results were analyzed across entire families rather than among genera or species.

RESULTS

Correlations were scaled between 0 and 1, with 1 representing perfect correlation. The postcanine tooth row lengths compared to the mandible lengths were 0.982 and 0.980 for the top and bottom, respectively. The postcanine tooth row lengths compared to the calvarium lengths were 0.869 and 0.863 for the top and bottom, and when compared to the cranial lengths, were 0.961 and 0.958. The mandible and calvarium

lengths were correlated at 0.898, the mandible and cranial lengths at 0.976, and the calvarium and cranial lengths at 0.969. All the traits were significantly correlated with each other.

Callitrichids showed significantly shorter maxillary and mandibular postcanine tooth rows relative to the three cranial measurements when compared to the other platyrrhine families (Figure 4; Table 2). *L. rosalia* displayed a significantly larger postcanine row relative to skull length when compared to other callitrichids. When compared to non-callitrichid platyrrhine families, however, *L. rosalia* exhibited a significantly smaller postcanine ratio relative to at least one skull measurement (Table 3). There were no significant differences between *C. pygmaea* and other callitrichids in their postcanine row relative to their skull lengths (Table 4).

DISCUSSION

Tooth row length's significant correlations with the mandible length, calvarium length, and cranial length, across families, confirm that a strong relationship exists between tooth row length and body size. Slightly lower correlations between the tooth row lengths and calvarium lengths may be due to the high quantity of missing calvarium data from the photograph measurements rather than to a weaker relationship.

My study demonstrates that the postcanine row of callitrichids is shorter relative to the various skull lengths when compared to non-callitrichid platyrrhines, indicating that their postcanine row consumes a smaller region of their skull. My study's results are consistent with those of the Plavcan and Gomez study. In interpreting these results, Plavcan and Gomez accepted Gould's assumption that phyletic dwarfism must be associated with ontogenetic scaling. In this instance, ontogenetic scaling was expected to manifest in the usual trend of negative allometry, or larger tooth area relative to decreased body size. Plavcan and Gomez, thus, could not assert callitrichids underwent phyletic dwarfism. Their study claimed that the unique callitrichid traits of higher twinning rates, claws, third molar loss, and no hypocones are not even suggestive of a dwarfing event or a series of such events. They argued that given the trend of third molar size reduction in non-dwarfed platyrrhines, the loss of this tooth in callitrichids may represent a simple continuation of a general trend unrelated to body size reduction. Plavcan and

Family Comparison	Upper Average to Mandible	Lower Average to Mandible	Upper Average to Calvarium	Lower Average to Calvarium	Upper Average to Cranium	Lower Average to Cranium
Atelidae-Aotidae	0.847	1.000	0.000	0.000	0.229	0.106
Callitrichidae-Aotidae	0.000	0.000	0.006	0.128	0.000	0.006
Cebidae-Aotidae	0.847	0.950	0.887	1.000	0.125	0.878
Pitheciidae-Aotidae	0.417	0.904	0.817	0.749	0.989	0.928
Callitrichidae-Atelidae	0.000	0.000	0.000	0.000	0.000	0.000
Cebidae-Atelidae	1.000	0.906	0.000	0.000	0.000	0.001
Pitheciidae-Atelidae	0.879	0.780	0.000	0.001	0.404	0.343
Cebidae-Callitrichidae	0.000	0.000	0.059	0.087	0.309	0.061
Pitheciidae-Callitrichidae	0.000	0.000	0.000	0.001	0.000	0.000
Pitheciidae-Cebidae	0.941	0.271	0.210	0.620	0.020	0.291

Family Comparison with Cebuella	Upper Average to Mandible	Lower Average to Mandible	Upper Average to Calvarium	Lower Average to Calvarium	Upper Average to Cranium	Lower Average to Cranium
Atelidae-Aotidae	0.913	1.000	0.000	0.000	0.300	0.145
Callitrichidae-Aotidae	0.000	0.000	0.017	0.307	0.000	0.014
Cebidae-Aotidae	0.912	0.950	0.941	1.000	0.170	0.934
Cebuella-Aotidae	0.000	0.000	0.164	0.342	0.024	0.110
Pitheciidae-Aotidae	0.516	0.904	0.891	0.836	0.997	0.966
Callitrichidae-Atelidae	0.000	0.000	0.000	0.000	0.000	0.000
Cebidae-Atelidae	1.000	0.906	0.000	0.000	0.000	0.002
Cebuella-Atelidae	0.000	0.000	0.000	0.000	0.000	0.000
Pitheciidae-Atelidae	0.935	0.780	0.001	0.002	0.501	0.433
Cebidae-Callitrichidae	0.000	0.000	0.131	0.249	0.458	0.117
Cebuella-Callitrichidae	1.000	1.000	1.000	0.994	0.995	0.988
Pitheciidae-Callitrichidae	0.000	0.000	0.000	0.006	0.000	0.000
Cebuella-Cebidae	0.000	0.000	0.467	0.321	0.656	0.348
Pitheciidae-Cebidae	0.974	0.271	0.280	0.722	0.029	0.374
Pitheciidae-Cebuella	0.001	0.001	0.018	0.038	0.005	0.015

Family Comparison with Leontopithecus	Upper Average to Mandible	Lower Average to Mandible	Upper Average to Calvarium	Lower Average to Calvarium	Upper Average to Cranium	Lower Average to Cranium
Atelidae-Aotidae	0.865	1.000	NA	NA	0.230	0.125
Callitrichidae-Aotidae	0.000	0.000	NA	NA	0.000	0.001
Cebidae-Aotidae	0.864	0.941	NA	NA	0.118	0.926
Leontopithecus-Aotidae	0.001	0.002	NA	NA	0.968	0.956
Pitheciidae-Aotidae	0.383	0.888	NA	NA	0.996	0.962
Callitrichidae-Atelidae	0.000	0.000	NA	NA	0.000	0.000
Cebidae-Atelidae	1.000	0.891	NA	NA	0.000	0.002
Leontopithecus-Atelidae	0.008	0.000	NA	NA	0.045	0.013
Pitheciidae-Atelidae	0.897	0.750	NA	NA	0.423	0.402
Cebidae-Callitrichidae	0.000	0.000	NA	NA	0.044	0.017
Leontopithecus-Callitrichidae	0.000	0.024	NA	NA	0.001	0.088
Pitheciidae-Callitrichidae	0.000	0.000	NA	NA	0.000	0.000
Leontopithecus-Cebidae	0.022	0.000	NA	NA	0.643	1.000
Pitheciidae-Cebidae	0.957	0.231	NA	NA	0.015	0.344
Pitheciidae-Leontopithecus	0.139	0.015	NA	NA	0.775	0.523

Table 2 (top). ANOVA for ratio values of specimens across families. Shaded cells are significant at $p < .05$.

Table 3 (middle). ANOVA for ratio values of specimens across families, including "family" *Cebuella*. Shaded cells are significant at $p < .05$.

Table 4 (bottom). ANOVA for ratio values of specimens across families, including "family" *Leontopithecus*. Shaded cells are significant at $p < .05$.

Gomez stated that while tooth size is determined early in ontogeny and seems less affected by systemic growth, body size reduction appears in truncating late growth. They suggested that callitrichids are likely under the effects of longer gestation periods rather than being dwarfed via ontogenetic scaling. Their work could not, on the other hand, disprove that callitrichids underwent body size reduction over time. Plavcan and Gomez claimed that their linear regression results would differ depending on whether the size reduction in callitrichids was rapid or gradual, and that the allometry of callitrichids is associated with a gradual decrease in body size over time.

Plavcan and Gomez reasoned that it was unlikely that cranium shortening and the subsequent lack of occlusal space caused the last molar to be pushed out of the postcanine tooth row. The work performed by Cai et al.¹⁷ also lends support to the rejection of Gould's hypothesis; it posited that tooth size and number might be regulated independently, and that changes in the number of molars can take place without affecting the molar area. There would be no selective benefit for the third molar to be forced out due to a lack of occlusal space, if the overall molar area does not necessarily change relative to the cranium and jaw.

My study, from the Plavcan and Gomez perspective, provides an even stronger argument for callitrichids not being phyletic dwarfs. *L. rosalia* and all other callitrichids in my sample show significantly smaller postcanine to cranial length ratios compared to non-callitrichid platyrrhines, reinforcing a lack of negative allometry and ontogenetic scaling in this family. Plavcan and Gomez showed that only *L. rosalia* consistently fell above the regression lines comparing tooth area to cranial size, which indicates that its postcanine row, compared to non-callitrichid platyrrhine species, was not significantly smaller relative to its cranium size. However, *L. rosalia*, being the largest callitrichid, still did not provide sufficient evidence to support Gould's dwarfism hypothesis.

The conclusions of Plavcan and Gomez, however, become questionable when considering the perspective of Montgomery and Mundy. Their 2013 study realized species-level mathematical analyses on gestation length, prenatal growth, prenatal growth rate, and age at sexual maturity, a measure of the length of postnatal growth and growth rate, in callitrichids. In contrast to what Plavcan and Gomez proposed, episodes of body mass reduction in callitrichids occurred concurrently with shifts in prenatal growth rate, and gestation length is not significantly shorter in callitrichids than in other primates, including non-callitrichid platyrrhines. Montgomery and Mundy also offered an alternative type of ontogenetic scaling that supported the dwarfism hypothesis. Using ancestral-state reconstruction, they discovered that the average rates of body mass reduction in callitrichids are somewhat lower than the well-studied episodes of island dwarfism in Pleistocene mammals. The timescale considered in these studies are typically in the orders of thousands of years, rather than millions. The percentage change in body size of callitrichids is therefore equal to or greater than most examples of dwarfism. The rate of change of adult body mass in callitrichids is like that of horses, which evolved over similar time lengths.

My study's additional comparison with *C. pygmaea*, the species that defies the Montgomery and Mundy trend in also possessing a lag in postnatal growth rate, suggests that its special ontogeny does not seem to inform its postcanine tooth row length. This raises a further research question as to whether callitrichid dentition and third molar disappearance share an underlying genetic link only with their stalled prenatal growth rate. These may not be influenced by the addition of a slower postnatal growth rate, as seen in *C. pygmaea*.

We do not know what caused the third molar in callitrichids to vanish, nor do we know why their bodies shrank over time. The area of callitrichids' first molar relative to the remainder of their molar row is consistent with the expected ratios from the proportion of genetic activators and inhibitors that regulate primate molar development. However, this does not explain their third molar's agenesis, or failure to grow in the embryonic stage. Genetic patterning mechanisms around third molar suppression are likely to be best understood by perusing the modulation of tooth patterns and associated genes around third molar agenesis. It is possible that the suppression and agenesis third molar may be achieved by altering existing and highly conserved genetic pathways.¹⁸

CONCLUSION

Moving forward, studies on extant human populations and other non-human primates might also elucidate the mechanisms of tooth agenesis. The failure to develop normal teeth is common in modern humans, typically affecting the teeth that develop last in each tooth class of the secondary dentition, the set of 32 permanent teeth that erupt in childhood and last until old age. The agenesis of the third molar is the most frequent. The greater susceptibility shown by the last developing teeth suggests an overall reduction of odontogenic potential. This could be produced by heterozygous loss of function mutations that reduce the gene dose during the final stages of tooth development.¹⁹ Thus, research on genetic patterning in callitrichids could have implications for understanding and correcting common agenesis-related mutations in humans.

To date, mutations in many genes have been identified in human families with tooth agenesis. Some of them, such as mutations in *MSX1* and *PAX9*, are commonly associated with reduced dimensions, shortened roots, and simplified form.¹¹ Whether the same genes are involved in the loss of the third molar in callitrichids is still a subject of research. Only one study has evaluated the *PAX9* gene in platyrrhines, in the genera *Callithrix*, *Saimiri*, and *Aotus* of the *Cebidae* family. The results obtained showed that these species share mutations in this gene that change three of the amino acids produced compared to humans and apes, except for *Aotus*, which creates two of these changed amino acids.²⁰ It is unclear at this point if this gene is having a similar effect in these cebid species as it is in humans. Further studies would need to discern any polymorphisms, sequence differences, or gene dose mutations in the *PAX9* gene of callitrichids and uncover if these are associated

with callitrichids' complete loss of their third molar. In fact, studies evaluating a larger number of genes known to regulate molar development are required to understand the specific molecular mechanisms responsible for dental variation within the platyrrhine clade. This could help clarify whether third molar agenesis is linked to genes known to affect body size.

Ultimately, my study supports an association between reduction in tooth number and phyletic dwarfism in callitrichids. Nevertheless, a lack of room in the cranium or mandible is unlikely to be responsible for an adaptive loss of their third molar. The disappearance of the third molar, then, is probably not a product of selection on a phenotype or dietary specialization. Rather, it seems to be related to shared genetic patterning effects underlying body mass, cranium length, and postcanine tooth row length.

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ASSESSMENT ON INTEROPERABILITY OF HEALTH INFORMATION EXCHANGES

BY VARUN NEIL WADHWA

ABSTRACT

This paper identifies several areas of improvement needed in health information technology, and makes four separate recommendations to achieve interoperability of Health Information Exchanges. **Keywords:** Functional Interoperability; Semantic Interoperability; Health Information Exchange; Health Information Technology; Electronic Medical Record

INTRODUCTION

Organic materials present in air tend to adsorb onto metal and form nanostructures to reduce the free energy between surfaces and the environment.¹⁶ A common model for these contaminants is based around Self-Assembled Monolayers (SAMs).¹⁷ These molecular structures are formed from exposure to fluids and gases with organic content and organize spontaneously into crystalline-like adatom structures. They commonly appear on gold and copper surfaces, both of which are common traps electrode materials.¹⁸⁻²⁷ Many SAMs are “alkanethiols,” characterized by single-bond hydrocarbon chains with sulfur heads.^{28]} hey can be removed by plasma cleaning.²⁹ These observations make them a candidate for the contamination leading to the elevated electric field noise observed in ion traps. This study investigates whether or not certain ion trap procedures can change the structure of organic surface contamination.

WHAT IS A HEALTH INFORMATION EXCHANGE? WHY IS IT IMPORTANT?

According to the American Health Information Management Association (AHIMA), “the primary function of a Health Information Exchange is to permit access to clinical information on demand at the point of care.” Indeed, as pointed out by Claudia Williams et. al.,³ in order for an HIE to be successful, it needs to be able to perform at least five basic tasks: 1) Electronically exchange laboratory results, 2) Electronically exchange care and discharge summaries, 3) Public health reporting, 4) Report on the quality of care thereby creating a feedback loop between the doctor and patient, and 5) Share information with patients.

Thus, a robust HIE would allow healthcare providers to access all relevant patient information on demand. Patient information includes everything that a healthcare provider needs in order to make medical decisions, such as medical history, current medical condition, pathology reports, CT scans, etc. In an ideal world, such a system would be accessible and updatable by healthcare providers, such as doctors, nurses, pharmacists, and medical technicians, whenever they come in contact with the patient. The patient’s medical history would, so to speak, follow him around throughout his lifetime, precluding the need for the patient to maintain the history on his own.

Health Information Exchange is also critical from the perspec-

tive of gathering data at the federal level in order to improve health outcomes, while increasing the efficiency and quality of healthcare delivery. An HIE should allow investigators to aggregate data at the city-, county-, state- or US-level in order to track the progression of disease and propose quick solutions. In one of the first meaningful steps towards interoperability, this office designed Certified Electronic Health Record Technology (CEH-RT), which established standards and other criteria for structured data that EHRs must meet in order to qualify for use in the Medicare and Medicaid EHR Incentive Programs. This gives assurance to purchasers and other users that an EHR system or module offers the necessary technological capability, functionality, and security to help them meet the meaningful use criteria.

CHALLENGES FACED BY HIES

Legal / Policy Issues:

HIEs and the transmittal of patient health information are currently governed by a patchwork of federal and state laws. In case of a conflict between similar federal and state laws, federal law is often pre-empted by state law.

State law specifies, among other things, 1) exactly what medical information to include in a patient's health record, 2) the circumstances under which a healthcare provider is allowed to release this information, 3) to whom the provider is allowed to disclose information, and 4) the purpose for which the information may be disclosed. In addition, state law requires providers to maintain patient medical records for several years (Table 1).

The potential for significant legal liability exists if a mistake is made in transmitting information between two HIEs/states. Legal algorithms would therefore need to be created to account for any contingency when transmitting information in this circumstance. To add to this complexity, these algorithms would have to be made for any combination of states, and would have to be modified any time the laws of a state are changed.

As pointed out by Claudia Williams et. al.,³ "...local networks have had to spend considerable time and legal resources crafting their own agreements." Since then, ONC has established the governance mechanisms and basic set of protocols governing the transfer of health information. States have taken an initiative to create HIEs, although their efforts have resulted in a mish-mash of public/private sector organizations tasked with operating them (Table 2).

Data Security:

Data security is, of course, critical to building trust in an HIE, especially given the multitude of recent hacking attacks on both U.S. government and private enterprises. On one hand, a perfectly interoperable healthcare system that can access the health records of every single U.S. resident would "improve the quality of health care, prevent medical errors, reduce health care costs, increase administrative efficiencies, decrease paperwork, and expand access to affordable health care." However, as the healthcare universe becomes more interconnected, it also becomes more vulnerable; and with private and potentially life-critical information on the line, quality cybersecurity

Retention requirement	# of states
Minimum 3-5 years	8
Minimum 6-9 years	15
Minimum 10 or more years	14
Dependent on type of provider	12
Dependent on patient condition*	2
TOTAL (incl. DC)	51

Table 1. States require healthcare providers to maintain patient records for several years. Number of states with specific record retention requirements. *Generally depends on whether patient is living or deceased. Data as of: 01/27/2016. Data source: <http://www.healthinfolaw.org/comparative-analysis/medical-record-retention-required-health-care-providers-50-state-comparison>.

ty is an essential part of any HIE. One protocol that may aid in this is the Direct Secure email protocol, as it offers end-to-end encryption rather than the server-to-server encryption offered by the Transport Layer Security protocol (the protocol currently used in most HIEs). Unfortunately, the Direct Secure protocol is largely unscalable, limiting its usefulness and preventing it from being used in systems requiring large HIEs.

Another security concern is that despite the fact that many federal agencies and industries specify a common set of protocols to authenticate patients requesting information from a central source, the healthcare industry has not done so. Not only does this lead to potential identity problems and data hazards, but it also raises the costs of doing business for all members of an exchange.

Entrenched Interests / Monopolization of Information:

Another key challenge faced by HIEs is entrenched interests that see HIEs as a threat to their business model. Indeed, hospitals perceive patient data “as a key strategic asset, tying physicians and patients to their organization.” (Grossman et al.⁴). For-profit hospitals are substantially less likely to share data (Adler-Milstein et al.⁵).

Epic Systems Corporation,⁶ a privately held company, is the largest provider of healthcare software in the world and exercises the same power as phone carriers to force subscribers to stay. According to the company’s website, 190 million people around the world (or around 2.6% of the global population) have an electronic record with Epic. Within the US, Epic holds medical records of well over half the population. Although they do offer tools for HIEs (e.g. Care Everywhere and the PHR Lucy), Epic still monopolizes the market for EMR storage by limiting exchanges via high costs and restrictions. The company appears to be especially loath to exchange health information with other EMR vendors who are direct competitors.

Another important issue is the lack of separation between regulators and the companies they are regulating; for instance, Epic’s founder, Judith R. Faulkner, is a member of the Health IT Policy Committee at ONCHIT. This clearly creates conflicts of interest between the objectives of her company and her government position.

Costs:

The upfront cost of licensing, installing, and using EMR data systems can be in the hundreds of millions of dollars for a large hospital system. Additionally, operational costs can be high. There is a misperception in the medical community that healthcare providers will pay these private HIE costs, even as the benefits of being able to access medical information anywhere will accrue to the payer (patient/insurance company).

As aforementioned, virtually all hospitals have some system of EMRs in place and so this is largely a problem of the past. However, the high cost of EMR implementation and regulation puts disproportionate burden on smaller providers of healthcare, who are thus likely to be especially resistant to change. In small multi-physician practices, it’s estimated that the EMR in-

stallation team will require more than 600 hours to implement the EMR system, and on top of that it will require an average of 134 hours for each physician to become familiar enough with the system to use it effectively with patients. During this EMR implementation period, which can last longer than a year, it is generally anticipated that the practice will see up to 50% fewer patients.

Furthermore, even once the EMR system has been implemented, these small practices often see a decrease in productivity and therefore a decrease in revenue. Put simply by Dr. John Haughton, the chief medical informatics officer at Co-visint, “If you’re putting in a new [EHR] system and it takes you 1 minute more per patient, that half hour [per day] is two billing slots. If you’re getting \$74 for a patient visit, then it’s \$150/day times, maybe, 200 days, your office is open. That’s \$30,000 per year.” These cost barriers are one of the main deterrents to the spread and effectiveness of private HIEs.

Public HIEs, on the other hand, have been funded by state governments in the hope that they will bring costs down for everyone; unfortunately, their own cost structure was deemed too high, especially in view of competing priorities. Public HIEs are therefore not self-sufficient and generally rely on state aid for operational expenses.

Data Quality And Interoperability:

According to a recent report on Healthcare IT by JPMorgan, “[Health Information Technology] buyers emphasize the need for higher quality data...over increased data volume, followed by a need for interpretation and interoperability.”

It is worth noting at this point that there are two kinds of interoperability: functional and semantic. Functional interoperability gives healthcare providers read-and-write access to patient records, regardless of which EMR vendor is storing the records. Complete functional interoperability, however, is unlikely to be implemented. This form of interoperability suggests that all medical data is shared fully between EMR vendors, effectively forcing them to offer a commoditized product. All EMR vendor features/functions become uniform in such a scenario. Different healthcare providers, however, may want different features based on their medical preferences. As an example, relate this to the car industry. If there was complete functional interoperability between car companies, all cars sold would be nearly identical. These cars would therefore not appeal to many customers because everyone has a different taste in cars. Many car companies would collapse because there would be no need to sell the exact same product, reducing competition and the need to innovate. Thus, functional interoperability may not be a desirable feature of HIEs.

Semantic interoperability, on the other hand, is the ability to share and interpret coded data. In semantic interoperability, the codes used for the content of the message and the format/order of these codes have to align with the standards of both vendors. In other words, the meaning of EMR content is preserved in a semantically interoperable exchange between two EMR vendors.

Currently, one EMR vendor cannot interpret another EMR

HIE run by:	# of states
Newly created Public or Private entity	22
State govt contracts HIE functions to an existing HIO*	11
State government runs the HIE via an existing agency (e.g. Dept of Health)	10
Partnership between an existing state agency and a newly created organization	4
No recent information on operating structure available	2
Uses an alternate state's system**	1
TOTAL	50

Table 2. Although states have taken the initiative to create HIEs, the effort has resulted in a mish-mash of public/private sector organizations tasked with operating them. Number of states utilizing various public/private structures to run HIEs; data as of Dec-2013. *HIO: Health information organization. **CT uses the RI HIE system. Data source: <http://www.healthinfolaw.org/comparative-analysis/status-health-information-exchanges-50-state-comparison>.

vendors' data when information is shared. Even more worrying, two different institutions (e.g. two hospitals) with the same EMR vendor cannot interpret each other's information. Sometimes, this is by design: hospitals prefer a lack of semantic interoperability because they view their data as a competitive advantage and want to make it difficult for other hospitals to see it. Additionally, they may not trust data from other hospitals. Finally, hospitals don't want external data to appear as if it came from their own (internal) sources because this has the potential to create confusion. Other healthcare providers, however, may prefer semantic interoperability to reduce the burden of interpreting information. Thus, among HIE stakeholders, there is much less pushback to implement semantic interoperability than there is to implement full functional operability.

Interoperability Standards:

Since each EMR vendor has established its own data model, it's not feasible to compel them to change their entire design to be semantically interoperable with other vendors' data systems. Instead, writing interfaces to connect two models together works as a better solution in the short run.

There are many different standards for HIE's, and many more being worked on. A popular choice that is used at the local and national level of the US are the Health Level 7 Standards. Health Level 7 is an international organization composed of health information experts, created specifically to develop standards for hospitals information systems. The first version of their standards was the HL7 v2, which was released in 1987 and widely adopted in North America. Unfortunately, it did not scale well, and thus, in 1995, HL7 v3 was developed. HL7 v3 is considered to be incredibly complex, with many other drawbacks that have been discovered regarding its interoperability and its ability to be implemented. And despite being adopted in countries such as the UK, Canada, and Germany, it was

not selected for use in the HITECH act in the US. This, along with the other aforementioned factors, was the driving force behind HL7's most recent HIE standard, the Fast Healthcare Interoperability Resources (FHIR). The FHIR is designed to complement the best parts of HL7 v2 and v3, as well as already existing Internet Standards. Its goal is to be easily consumable by reducing the complexity of implementation and usage.

Once an HL7 interface is built, the HL7 Clinical Data Architecture (CDA) sets standards between EMR vendors. The data can then be interpreted by machines of various EMR vendors. While these interfaces can be time-consuming and costly to establish, they offer significant cost savings in the long run and are generally preferable to the alternative – sending the file as a read-only PDF/fax.

Interoperability Solutions:

Internationally many other countries have their own HIE systems in place, implemented with varying degrees of success. The French electronic healthcare record system, the DMP, has been predominantly successful in large part due to cooperation of healthcare vendors and the acceptance of the French interoperability framework (CI-SIS). The free service makes information required for patient care more easily available, and helps to facilitate the communication between patients and healthcare professionals.

On the other side of the world in Australia, there have also been large strides towards interoperability, namely the Personally Controlled Electronic Health Record (PCEHR). The PCEHR is a shared electronic health record set up by the Australian government, designed to provide a secure summary of people's medical history. In July of 2016, the Australian government opened the Australian Digital Health Agency whose primary responsibility is to oversee the PCEHR system, as well as being respon-

sible for all other national digital health services and systems.

Here in the U.S., the drive towards interoperability and successful HIE has come more from the private sector. One of the leaders of this is the Sequoia project, which is an independent advocate of nationwide health information exchange. Their signature initiative, eHealth exchange, currently spans all 50 states, is in use in over 65% of U.S. hospitals and is easily the largest health data-sharing network in the US, bringing coverage to over 100 million patients. By leveraging a common set of standards, legal agreement and governance, eHealth Exchange participants are able to securely share health information with each other, without additional customization and one-off legal agreements. While there is still a long way to go and many obstacles to overcome, there is positive progress being made everywhere in the world.

CONCLUSION / POSSIBLE SOLUTION

The technology to electronically exchange information securely has existed for some time now. Indeed, the March 2012 paper by Claudia Williams et. al.³ cited before says that "...the building blocks required to initiate all three forms of exchange are complete, tested, and available today. These standards are already in use by private networks and electronic health record vendors to exchange documents within their own networks."

Many healthcare providers, however, prohibit other companies from accessing their information, known as "information blocking." By refusing to transfer patient data, they effectively reduce competition and also claim to avoid legal issues with data security. Some practices of information blocking include "excessive costs for information-sharing; a lack of contract transparency for technology buyers, who may not appreciate the costs of sharing information; efforts by vendors to make it difficult to download information from EHRs and port their systems to competitors' systems; and collusion between providers and vendors to prevent information from following to other software systems or healthcare providers."

Further, even if companies were willing to share their data, the current EHRs largely lack the design and features for interoperability. A possible solution that has been proposed is to focus on the "high-value use cases, such as transitions of care, outcomes measurement, and public-health reporting". Build the interstates and highways first, and the smaller roads will follow.

The government has little power to force full functional interoperability because strong policy responses would be disruptive and expensive, and would result in significant pushback from larger healthcare providers/EMR vendors. However for real change to be possible the government has to work alongside the private organizations setting the industry standards, and needs to have a concerted push for standards yielding end-to-end interoperability. They have done this to some extent, creating the CDA for the HL7 interface to translate information for HIE standardization which was a good starting step towards semantic interoperability.

Many private corporations have already achieved semantic interoperability to a limited extent. Surescripts, for instance, is a company that moves national drug codes (NDCs) between

pharmacies, doctors' offices, and insurance companies. Their systems can therefore interpret drug data because the codes are standardized. CPT-4 codes (Current Procedural Terminology owned by the American Medical Association) are another example of this standardization. CommonWell, a joint venture by multiple EMR vendors (led by Cerner), is currently also working towards solving the general issue of interoperability, and are one of the leading groups experimenting with the new FHIR. They believe that users and providers should be able to access medical information no matter where care occurs. CareEquality (led by Epic), CommonWell's rival, also focuses on this issue.

Given the significant progress made by government and private corporations in standardization to achieve semantic interoperability, interpretation of medical data will eventually be achieved. Functional interoperability, however, is unlikely to be implemented, due to resistance from virtually all players in the industry.

The HL7 interface works well in many cases, but is often too expensive and complex for use by the everyday doctor. The federal government should therefore consider initiating the following policy responses:

- 1) Compel EMR vendors to follow a specific format when transmitting information, going forward. For instance, the federal government could mandate the use of NDC and CPT-4 codes in EMRs.
- 2) Consolidate power at the federal level on HIEs. Each state currently has its own standards for health information exchanges, which limits the extent of semantic interoperability. Giving this power to a single source will help create standardization.
- 3) Pass laws to ensure that HIEs are not held responsible for security and privacy failures unless egregious misconduct can be demonstrated. For example, mandating arbitration in cases of disputes will help the industry overcome legal concerns. EMR vendors would be more likely to adopt interoperability if legal issues were resolved ahead of time.
- 4) Ensure that the roles of regulator and regulated are clearly delineated. While having industry participants on the boards of regulators can be beneficial, it also has the potential to create serious conflicts of interest.

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